

Preface

This manual provides the information needed to carry out maintenance on the WRO 300 / WRO 300 H and to identify and remedy any fault that may occur.

In this manual general information about the WRO 300 / WRO 300 H is stated as WRO. The complete model name WRO 300 or WRO 300 H is only used for specific model information.

Publication no.	HCEN12073 rev 05.2006
Denomination	Service manual



CAUTION

Dialysis machines that are supplied with water from WRO, must comply with IEC 60601-2-16.

Definitions of expressions in the manual



WARNING

Is used to alert the user/operator **not to take** a certain action, which if taken can cause a potential hazard and result in a serious adverse reaction, injury or death. A warning may also be used to alert the user/operator **to take** a certain action to avoid the potential hazard as above.



CAUTION

Is used to alert the user/operator **to take** a certain action to protect against a potential hazard which, if ignored, could have an adverse effect on the patient or the device. A caution may also be used to alert the user/operator **not to take** a certain action to avoid the potential hazard as above.

NOTE

A reminder to the user/operator on normal treatment activity and on what is a suitable action in a particular situation.

Safety considerations



WARNING

Unauthorized installation, modifications, alterations or repair of the WRO may result in malfunctioning or have other serious consequences for the safe operation of the equipment.



CAUTION

1. The WRO, may only be operated by persons trained in this equipment and who have studied the instructions in the Operator's Manual. If the WRO does not perform as described in the Operator's Manual, it should not be used until the condition is rectified.
2. The operator should pay attention to alarms and follow the instructions, warnings, cautions and notes given in the manual.
3. The use of mobile telephones or communication equipment in the vicinity of the WRO could adversely influence the performance of the machine.
4. WRO needs special precautions regarding EMC and needs to be installed and put into service according to the EMC information provided in the operators and service manual.
5. The WRO, will perform as designed only if it is used and maintained in accordance with Gambro's instructions. Any warranties made by Gambro with respect to the WRO, are void if the equipment is not used in accordance with the instructions provided. Gambro will not accept responsibility for any damage or injury resulting from improper use or maintenance or unauthorized repair.
6. The user must verify the quality of the protective earth in the installation.
7. For USA: US Federal law restricts this device to sale by or on the order of a physician

NOTE

During transportation and storage the equipment must be kept in its original packing. If transportation or storage time is more than 15 weeks the environmental data relating to the operation must be fulfilled.

CE-marking

This product is CE marked in accordance with the requirements in EC Council Directive 93/42/EEC of 14 June 1993 concerning medical devices.

Notified body is: British Standards Institution (BSI) with the notified body number 0086.

The symbol below means: Year of manufacturing.



The CE-marking of this manual is only valid if the device which it describes is CE-marked.

Manufacturer

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1 Introduction

1.1 Intended use

The Gambro WRO water purification unit is designed to be used as a dialysis accessory device to produce water used to prepare and dilute dialysis concentrate to form dialysis fluid by using the reverse osmosis concept. It is intended for use in conjunction with one dialysis machine, provided that the input flow and pressure demands correspond to the output of the WRO unit.

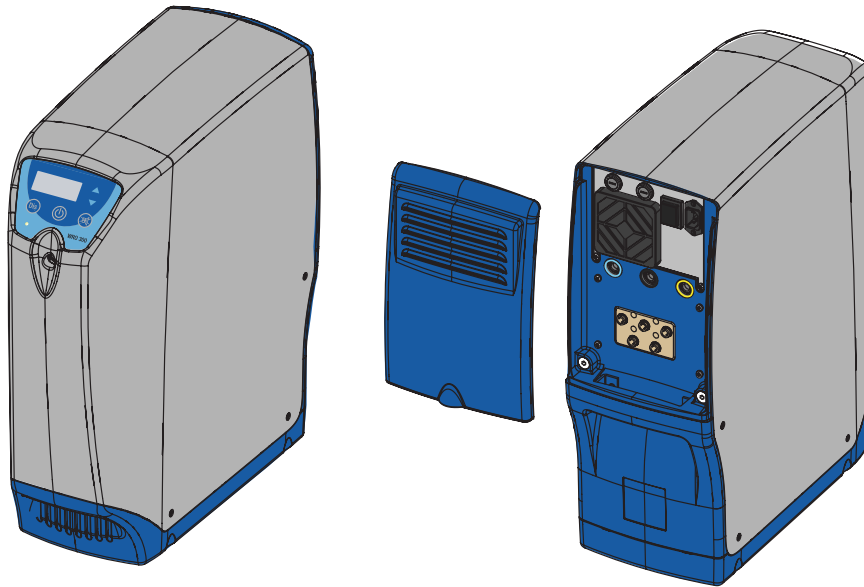


Fig.1.1 Views of the WRO 300.

1.2 Description of reverse osmosis

Reverse osmosis is a membrane filtration process and is the most widely used technique for the purification of water for dialysis. The critical part of a reverse osmosis system is the semipermeable membrane, i.e. a membrane which allows the passage of water but retains most of the dissolved salts, particles, bacteria and pyrogens.

When the water is brought into contact with such a membrane and a high pressure is applied, water will flow through the membrane (*Fig.1.2*) but most of the other constituents will remain on the original side of the membrane and be flushed to drain.

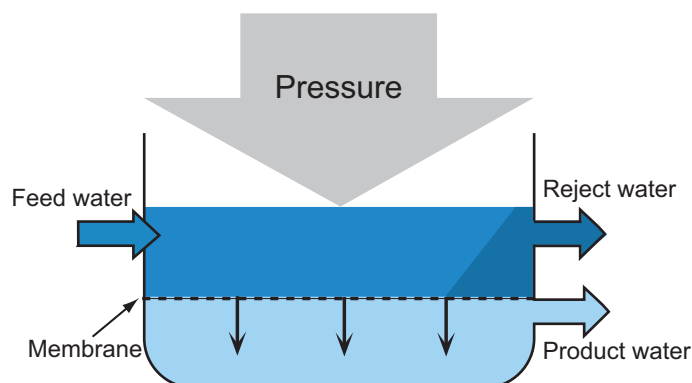


Fig.1.2 Principal of reverse osmosis

1.3 General function

1.3.1 Water purification

Tap water contains many types of contaminants such as dissolved inorganics and organics, particles, bacteria and pyrogens.

A dialysis patient is exposed to very large amounts of dialysis fluid that come into close contact with the patient's blood. A large number of substances present in tap water have been proven to have a negative effect on the dialysis patient. It is therefore evident that much higher standards must be set for water used in the preparation of dialysis fluid than for tap water. Water used for dialysis must therefore be purified.

The principal flow diagram is shown in [Fig.1.3](#).

A high-pressure pump feeds the pretreated water into the RO module. The pressure forces part of the water through the membrane where most of the contaminants are retained. The product water (permeate) then leaves the RO module through the product water outlet. The rest of the water containing the retained contaminants (reject) leaves the RO module through the reject water outlet and is then diverted to drain or back to the tank. The percentage value set for the water save function is how much of the reject water that goes back to tank. Excess product water and part of the reject water are recirculated to further reduce water consumption and to ensure high flow velocity over the membrane surface.

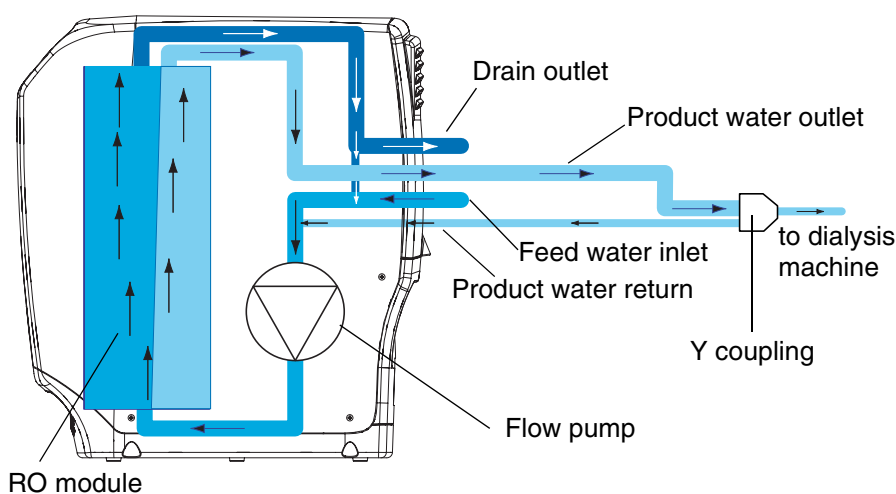


Fig.1.3 Principal of WRO 300

Pretreatment

An acceptable quality of the feed water is required, i.e. in accordance to ["Technical data" on page 5:1](#). The feed water is usually pretreated with such as active carbon filters, softener, and particle filters before it is supplied to the WRO. Depending on the local water quality and regulations different pretreatment equipment may be required.

1.4 Water quality

1.4.1 Microbiology

The microbiological quality of the dialysis fluid is a function of a number of factors. Neglecting only one of these could result in a poor water quality.

Example of these factors are

- Adequate feed water quality
- Regular and pro-active disinfection of the WRO
- Correct arrangement of the drain lines, i.e. air gap to avoid microbiological contamination from the drainage system.

Chemical quality

To get an indication of the water quality, the WRO measures the conductivity of the water (expressed in microsiemens per cm, $\mu\text{S}/\text{cm}$). The conductivity measures of the amount of dissolved salts in the water. The conductivity should mainly be used as an indicator of the performance of the reverse osmosis system.

!

CAUTION

The conductivity value does not always indicate the suitability of the water for dialysis. For example aluminium may be present in concentrations well exceeding recommended standards without affecting the conductivity.

Therefore the quality of the water, used for dialysis, should be verified by regular water analyses.

NOTE

The regulations and guidelines on the quality requirements for dialysis water varies in different countries around the world.

The EUROPEAN PHARMACOPOEIA has issued rules (1992) that regulates the microbiological as well as the chemical requirements for water for dialysis in Europe, as well as analysis methods to be used both for the microbiological and chemical testing of water.

2 Service & Technicians guide

2.1 Overview

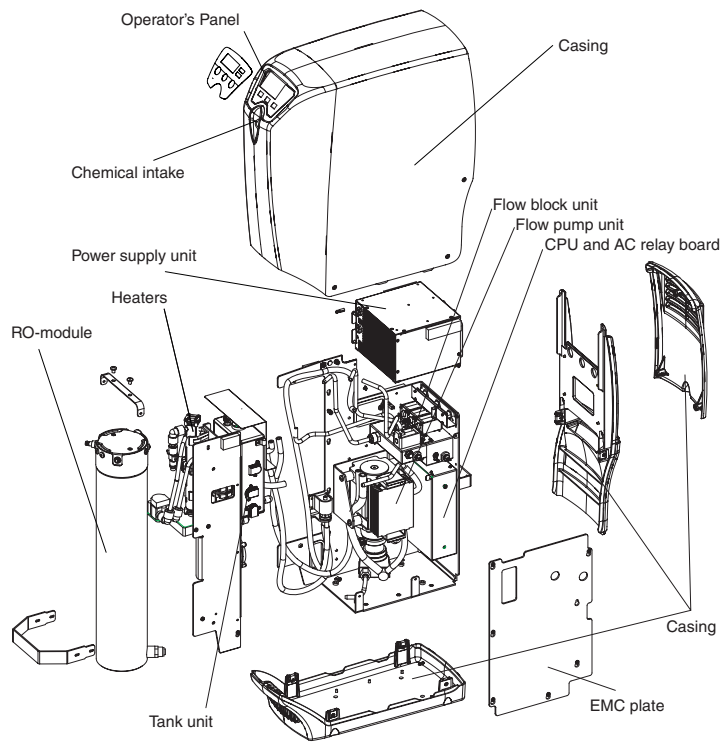


Fig.2.1 WRO 300 H overview

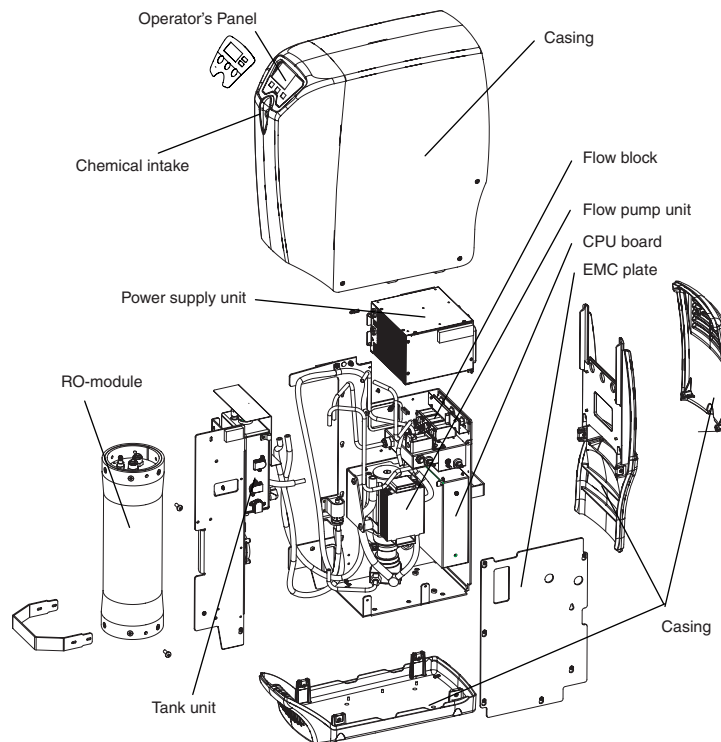


Fig.2.2 WRO 300 overview

2.2 Flow diagram

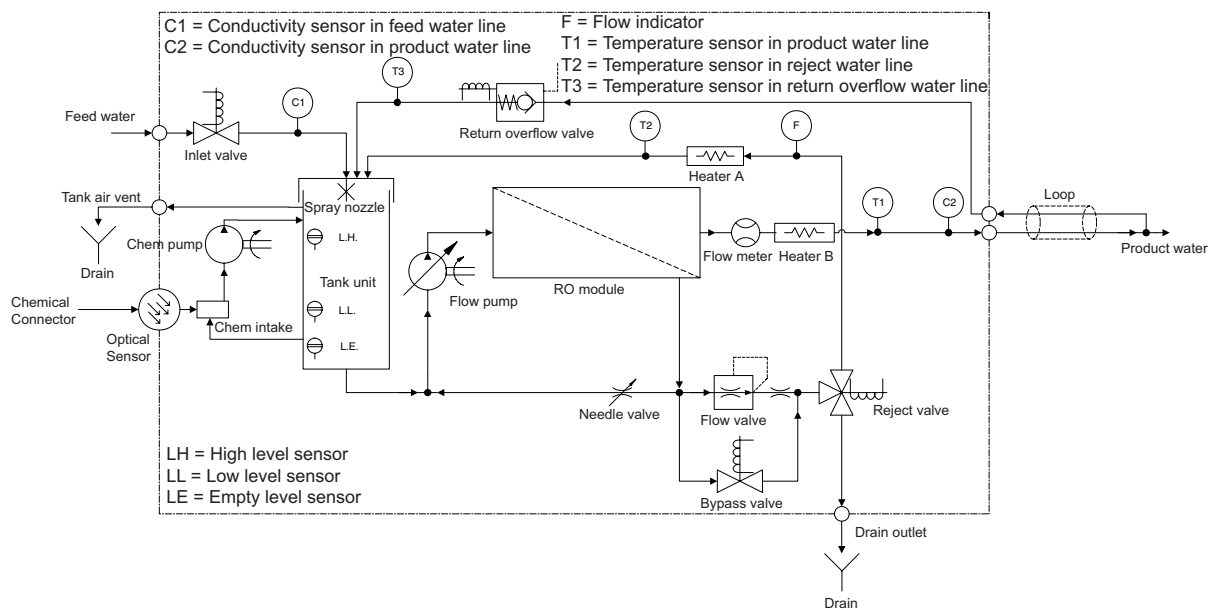


Fig.2.3 Flow diagram for WR 300 H

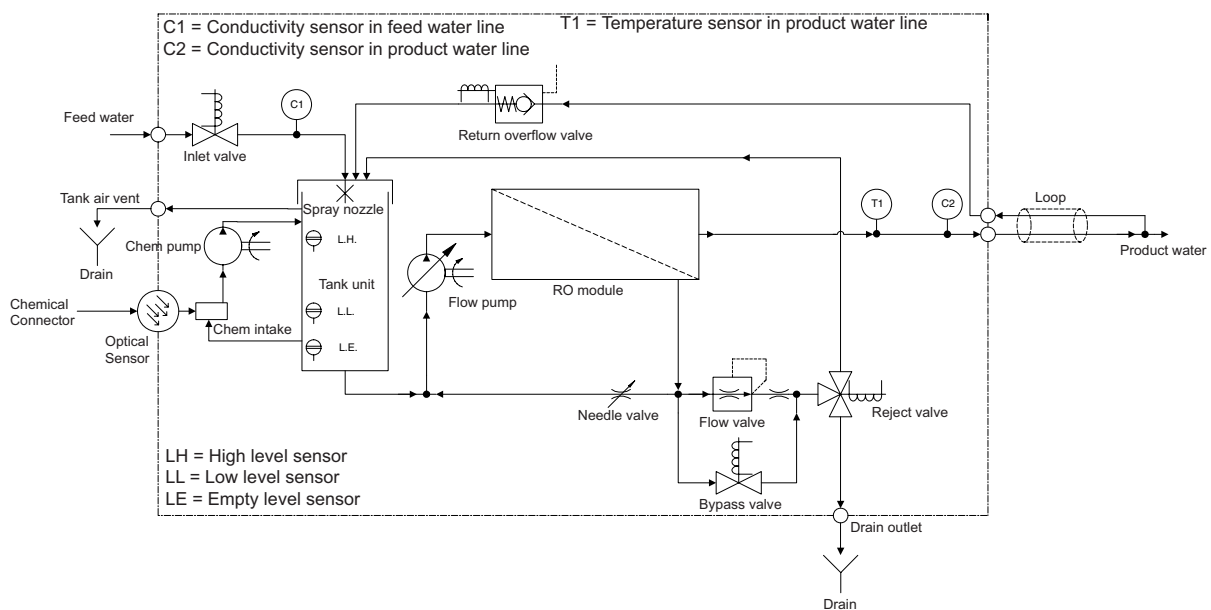


Fig.2.4 Flow diagram for WRO 300

2.3 Functional description

2.3.1 RUN mode

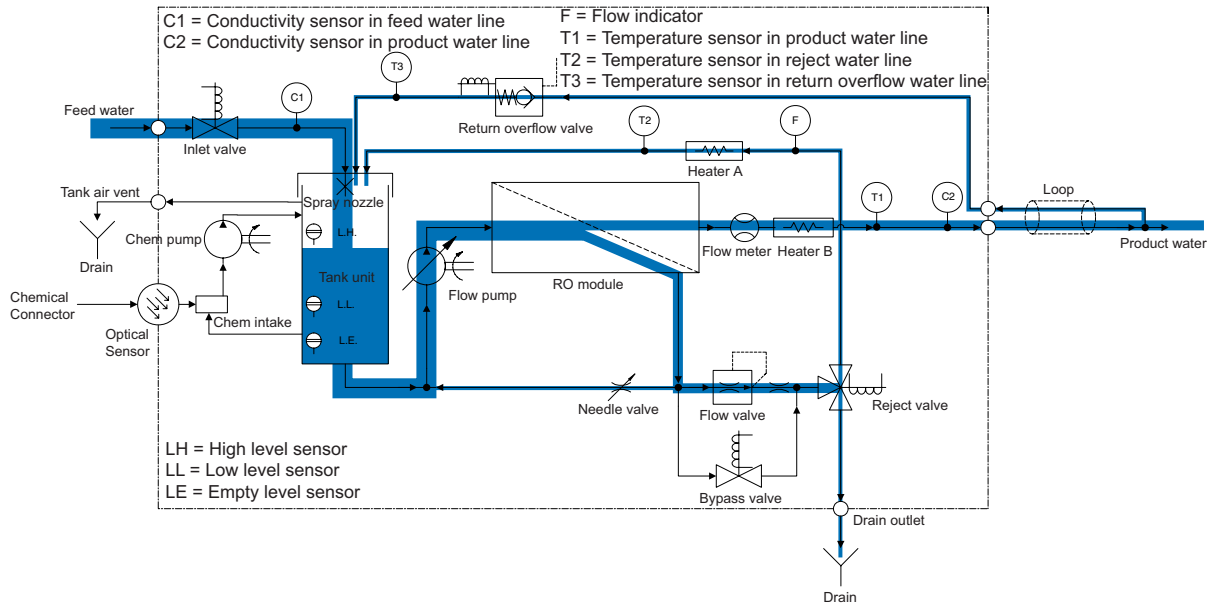


Fig.2.5 Flow during operation

Initiate

The WRO is started when  is pressed for three seconds. In WRO 300 H a temperature check (T2) in the reject water line is always performed before starting in RUN mode. If the temperature is above 45 °C a cooldown is performed in the same way as Active Cooldown, refer to [“Active cooldown” on page 2:6](#).

The inlet valve opens to enter feed water into the tank. When the water level is above the Low level sensor (LL) the Flow pump starts, the speed ramps up during the initial 2-3 seconds, and feeds the water to the RO module.

The product water is distributed from the RO module through the product water loop. Excess water not consumed by the dialysis machine will return to the tank via the Return overflow valve. The pressure in the product water loop depends on the amount of return flow.

The reject water from the RO module is partly diverted to the Reject valve through the Flow valve and partly back to the Flow pump through a needle valve.

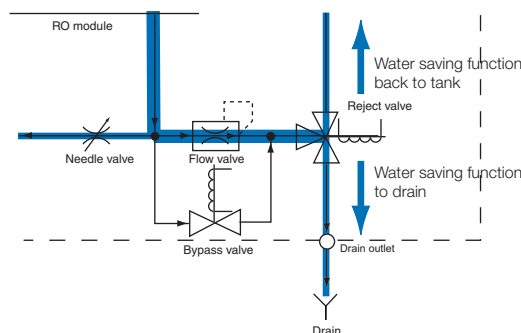
Auto flush

During Standby an Auto flush is regularly performed, if enabled in preset, within specified intervals to exchange the water in the WRO. The Auto flush circulates the water in the WRO within specified preset intervals, some water goes to drain and the inlet valve opens to enter water into the tank. The preset of Auto flush in GXP software is done by setting two time values. One to specify the time between the Flushes and the other for how long the Flush should last. Refer to [“Auto flush” on page 2:41](#).

Water save function

The three-way Reject valve alternates continuously to be opened to drain or open to tank. Water saving functionality can be obtained by a percentage value being set. This determines how much of the time the Reject valve is open to the tank. The settings of the Water saving function can be done in the software GXP. Refer to [“Water save” on page 2:35](#).

During the initial two minutes in the start up phase the Water save function is disabled and all reject water goes to drain.



Water level in tank

During RUN mode the inlet valve opens when the level is below Low level sensor (LL) allowing water to enter the tank. The inlet valve closes again when the water reaches the High level sensor (LH).

If no water is indicated by Empty level sensor, the Flow pump stops and  flashes fast.

If the Empty level sensor indicates no water at the same time as the Low level sensor indicates water, the Reject valve opens for five seconds back to the tank. The water flushes the Empty level sensor when it enters the tank to remove any air bubbles.

Stop (to Standby mode)

The WRO is stopped when  is pressed. The Bypass valve opens to release the pressure and the Reject valve is open to drain.

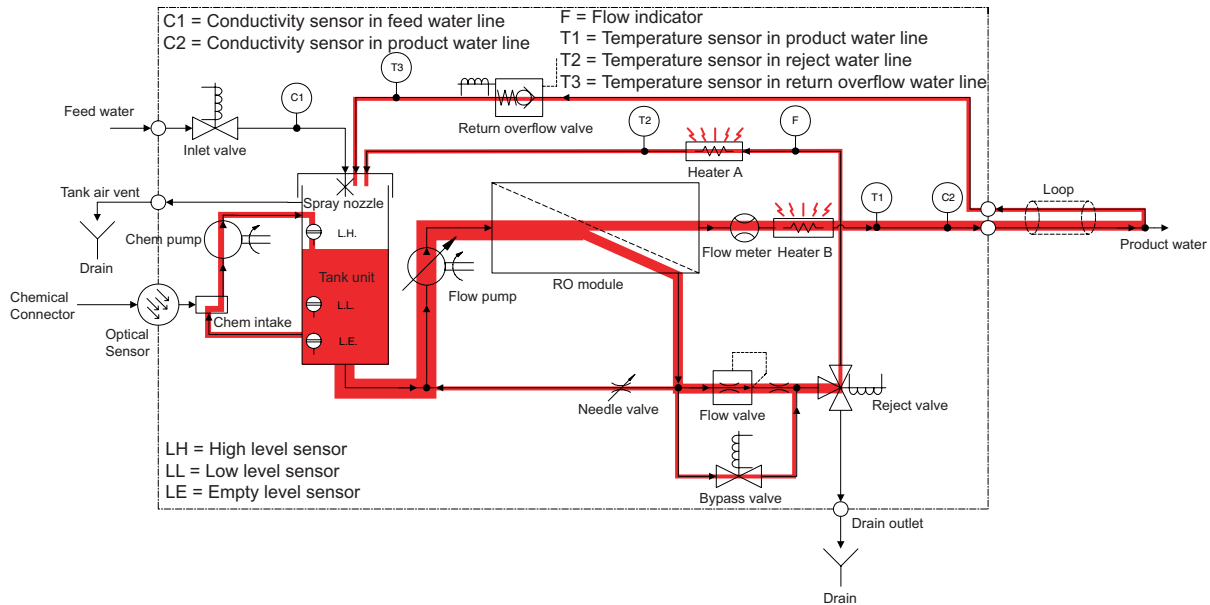
After ten seconds the inlet valve function stops to control the water level. The tank is emptied until the level is below the Low level sensor. Then the Flow pump stops and the WRO is in Standby mode.

Conductivity measurement

During RUN mode the product water conductivity is supervised (C1). The values from Conductivity sensors C1 and C2 are used to calculate rejection rate.

During the start-up phase a peak test is performed to confirm the function of the conductivity sensor in the product water line. During Standby mode or when the WRO has been turned off the salt concentration increases in the product water side in the RO module and this generates a peak in the conductivity value that is detected by the program.

2.3.2 Heat disinfection (only WRO 300 H)



Initiate

When **Dis** is pressed for three seconds **HEAT SELECT** is displayed. Select the Heat disinfection program with the arrow buttons and press **Dis** for three seconds.

The water in the tank is adjusted slightly above the Low level sensor. If no water enters the tank the alarm **302 INSUFFICIENT WATER SUPPLY** is issued.

Function check of temperature measurement

Both reject return (T2) and product (T1) water temperature amplifiers are checked for range and offset errors. The sensors are then checked for short or open circuit.

Function check of Heater protection

Next step is to test the heater control and flow supervision circuits. Both circuits are tested for power on and power off. Normal test time is about one minute. If stable flow cannot be reached within five minutes an alarm is issued. After this function check the Heat disinfection is started.

Start of Heat disinfection

The Flow pump starts with the Bypass and Return overflow valves opened, which make the system pressureless. The Heater A in the reject water line is activated followed by the Heater B in the product water line, which is activated when the flow is above 200 ml/min. The chemical pump is circulating the water in the chemical intake unit during the Heat disinfection.

The default preset value of the set temperature in reject and product water lines are 92 °C, refer to *"Heat Set temperature" on page 2:37*. After achieving 92 °C in the reject water line the Heat disinfection continues for 20 minutes.

During the Heat disinfection the water level in the tank increases with the temperature. If it reaches the High level sensor the Reject valve will open to drain for a few seconds to lower the level.

During Heat disinfection it is supposed to be no water consumption, otherwise the alarm **304 INCORRECT WATER LEVEL** is issued.

Active cooldown

The default preset is to run Active cooldown after the Heat disinfection is finished. The Flow pump circulates the water and the drain opens for three seconds. Then the inlet valve is open until the water reaches the High level sensor. After 20 seconds the temperature is checked and if it is below 45 °C the Active cooldown is finished. Otherwise the procedure is repeated until the temperature is below 45 °C. Then the WRO 300 H goes to Standby mode.

Cooldown

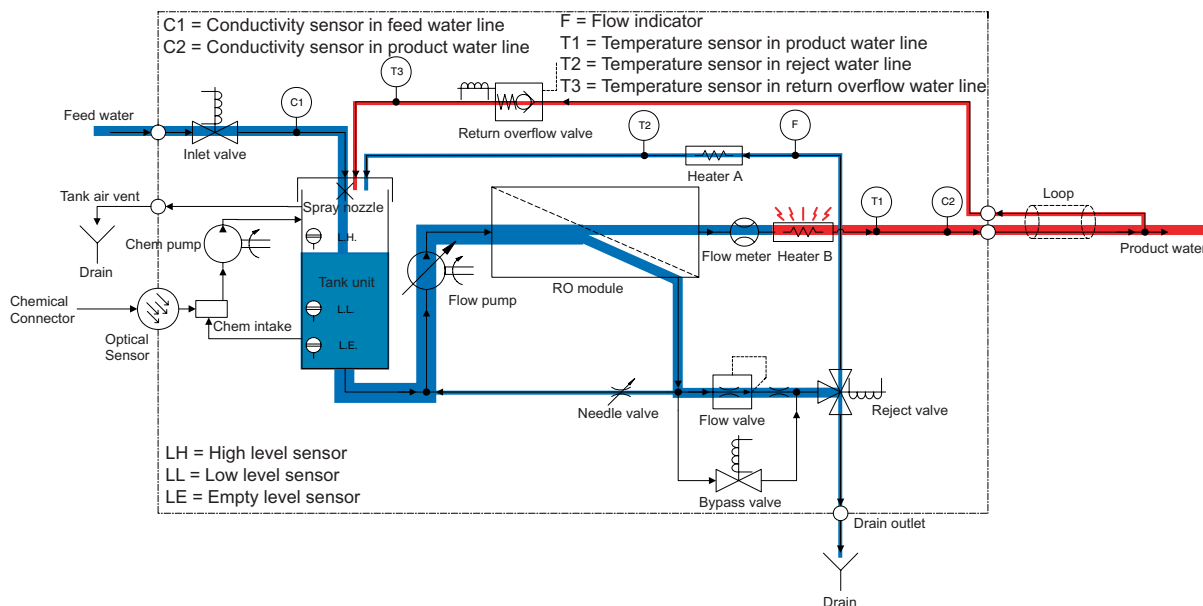
If Active cooldown is not selected the WRO 300 H goes to Standby and the temperature decreases slowly.

Stop during Heat disinfection

To stop the Heat disinfection press  for three seconds. An Active cooldown is performed, if selected. When the Active cooldown is finished the WRO goes to Standby mode.

If Active cooldown is not selected in the preset the WRO 300 H goes directly to Standby mode.

2.3.3 Low flow heat (only WRO 300 H)



Initiate

When **Dis** is pressed for three seconds **HEAT SELECT** is displayed. Select Low flow heat program with the arrow buttons and press **Dis** for three seconds.

The water in the tank is adjusted slightly above the Low level sensor. If no water enters the tank the alarm **302 INSUFFICIENT WATER SUPPLY** is issued.

Then the checks of temperature measurement and Heaters protection are performed, refer to [“Function check of temperature measurement” on page 2:5](#) and [“Function check of Heater protection” on page 2:5](#).

Start

The Flow pump and Heater B are turned on. The temperature is controlled to 83 °C (preset value, refer to [“Low flow heat” on page 2:38](#)) by adjusting the flow through the Heater B.

The product water flow depends on Mains Voltage (100, 115 or 220-240 V AC) and the temperature of the feed water.

Water level during Low flow heat

The water level in the tank will decrease during Low flow heat as the dialysis machine consumes water.

After eight seconds the inlet valve opens until the water reaches the High level sensor.

If the water level goes below the Low level sensor the Heater B is turned off and if it goes below the Empty level sensor the Flow pump is also stopped.

RO membrane protection

In the Low flow heat mode a pressure is created in the RO module. To avoid damage of the membrane the temperature has to be below 45 °C. If no water is consumed, hot water will return through the return overflow line to the tank. If the temperature in the reject water line (T2) reaches 40 °C the Low flow heat will turn off and the WRO goes to Standby mode.

2.3.4 Chemical disinfection and cleaning

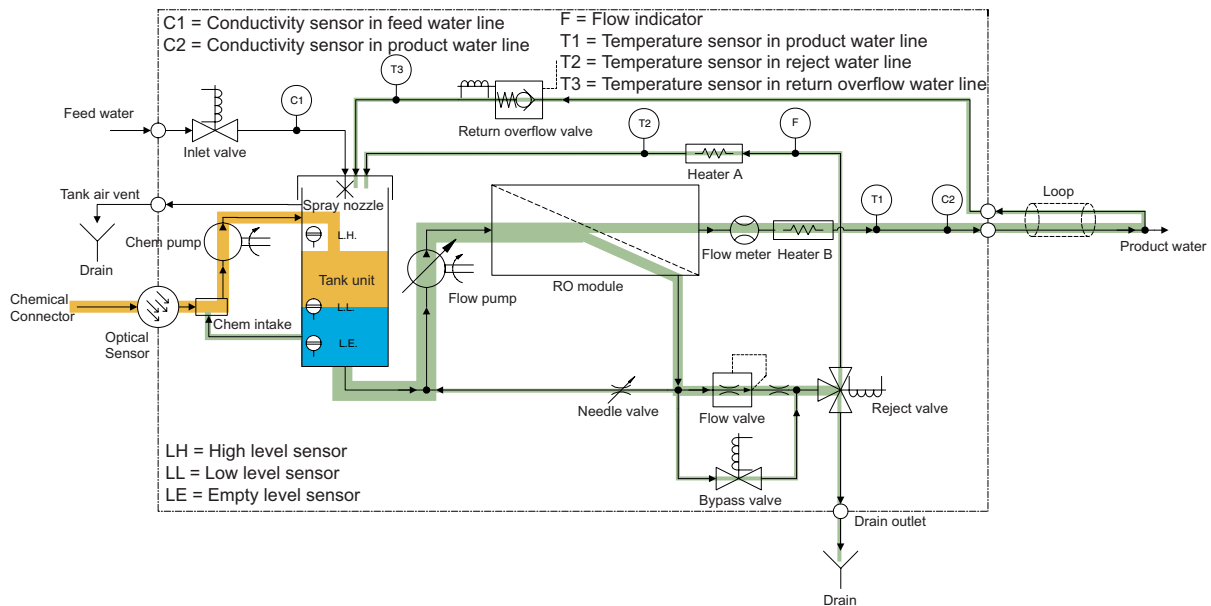


Fig.2.6 Flow during chemical disinfection or cleaning.

It is recommended to disconnect the dialysis machine from the WRO before performing Chemical disinfection or Cleaning.

CAUTION

If the WRO is not disconnected from the dialysis machine during the Chemical disinfection procedure, the test for residuals after the Chemical disinfection procedure must be performed on the dialysis machine according to the operator's manual of this device.

Initiate

An optical sensor detects when the disinfection wand is inserted into the chemical intake unit. The Flow pump stops the circulation if not the WRO already is in Standby mode. **CHEM SELECT** is displayed and **Dis** starts to flash (medium). Then the operator has to choose which protocol (presettled chemical procedures, refer to [“Disinfection protocol manager” on page 2:38](#)) to use. Select protocol to use and press **Dis** for three seconds.

Intake phase

In the program the Chem flag is set to TRUE and this internal setting will only change to FALSE when the procedure is completed.

The level in the tank is adjusted to be just above the Low level sensor. The Flow pump start and is running until the Low level sensor indicates air, then the Flow pump stops and the inlet valve opens and enters water until Low level sensor indicates water again.

The Chemical pump runs until the presettled amount of chemical solution has entered into the tank. During the chemical intake the Flow pump circulates the fluid in the flow path. When the chemical intake is finished the Flow and Chemical pump stop the circulation in the WRO and **401 REMOVE WAND** is displayed.

After ten seconds and when the disinfection wand is disconnected the tank is filled up to the High level sensor. In chemical disinfection mode the bypass valve is closed, but in Cleaning mode the bypass valve is open. The Flow pump starts the circulation through the RO module, the chemical pump starts the circulation through the chemical intake unit and

Return overflow valve opens. The circulation in the flow path will continue for one minute. Then the speed of the Flow pump is reduced and the bypass valve opens (only Chemical disinfection) and Reject valve opens to drain. Both valves closes and both pumps stops when the level is below the Low level sensor.

If the procedure is interrupted the WRO always remember that the chemical procedure has been started. A rinse is always required after an interruption. *RINSE* is displayed and the alarm **404 RINSE REQUIRED** appears. The rinse must be completed before the Chem flag is set to FALSE.

Dwell period

If the intermittent Run is disabled (in the preset, refer to *“Default setting of disinfection protocols” on page 2:40*) no circulation will be performed in the flow path.

If the intermittent Run is enabled (in the preset) circulation will be performed in the flow path according to the settings of circulation time and stop time. During dwell period the procedure can be interrupted by pressing  for three seconds and the WRO continues to rinse phase.

Rinse

The rinse phase consists of three parts that is repeated until the presetted rinse time is fulfilled. The three steps are.

- The Flow pump start at low speed. Bypass valve opens and Reject valve opens to drain. The Flow pump stops and the valves closes when the level is below Empty level sensor. Then the inlet valve is open until the water reach High level sensor. Then these steps are repeated six times.
- The Flow pump circulate the fluid in the WRO for two minutes with Return overflow valve open and Reject valve open to tank. The Chemical pump is also running.
- The Flow pump and valves closes and the circulation stops for one minute.

During the final five minutes of the rinse the conductivity value is checked in the Product water line. If the value is below the notification limit the rinse will be finished within the presetted rinse time. Otherwise the rinse will be extended with 30 minutes. As soon as the conductivity value is below the notification limit the extended rinse procedure will be finished. If the conductivity value still is above the notification limit after 30 minutes the alarm **403 HIGH CONDUCTIVITY RINSE** will be displayed.

2.3.5 Preservation with freeze protection

The intake of the preservation solution is done through the Feed water inlet. An external pump is required to enter the solution into the tank.

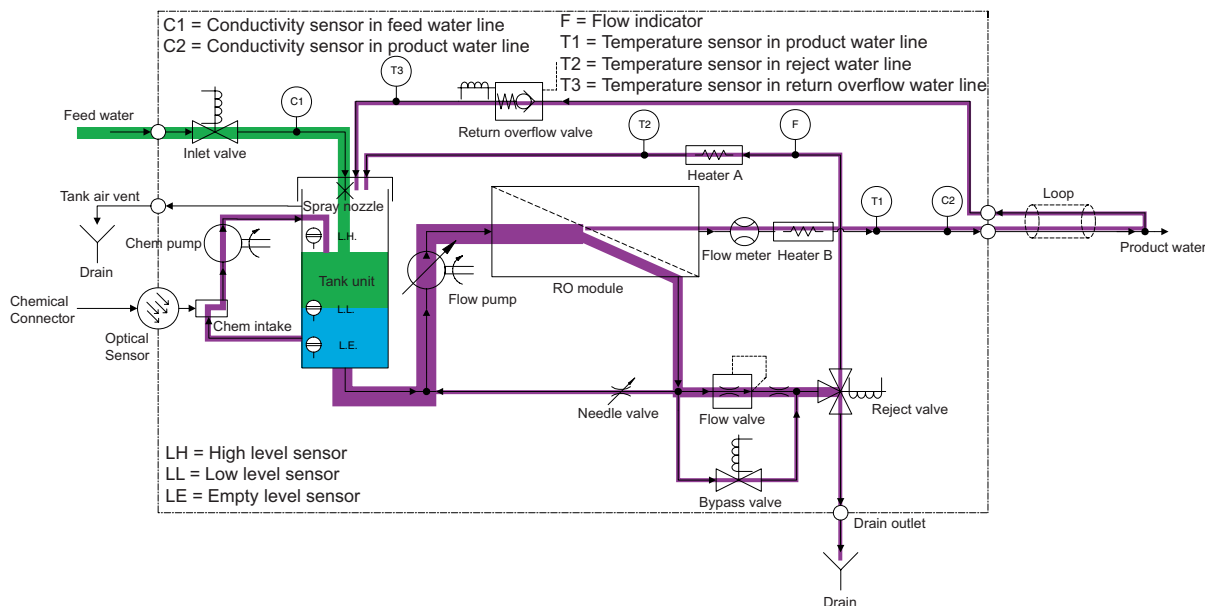


Fig.2.7 Flow diagram during Preservation mode.

An optical sensor detects when the disinfection wand is inserted into the chemical intake unit. The Flow pump stops the circulation if not the WRO already is in Standby mode. **CHEM SELECT** is displayed and **Dis** starts to flash. Then the operator has to choose a protocol for preservation (presettled chemical procedures). Select protocol to use and press **Dis** for three seconds.

NOTE

The disinfection wand is only inserted to initiate the **CHEM SELECT**. The preservation solution enters via the Feed water inlet.

The intake phase starts by pressing **Dis** for three seconds and **INTAKE** displays. The intake phase for preservation consist of three parts as below.

- The inlet valve is open until the water reach the High level sensor. The Flow pump start at low speed. Bypass valve opens and Reject valve opens to drain. The Flow pump stops and the valve closes when the level is below Empty level sensor. These steps are repeated six times and then **501 REMOVE WAND** is displayed.
- When the disinfection wand is disconnected the Flow pump circulate the fluid in the WRO for five minutes with the Bypass valve closed, Return overflow valve open and Reject valve opens to tank. The Chemical pump is also running.
- When the circulation is completed the Bypass valve opens and Reject valve opens to drain until the level in the tank is below Low level sensor.

When the preservation procedure is completed the alarm **404 RINSE REQUIRED** appears in the display.

2.3.6 Preservation without freeze protection

The function is identical with the *“Intake phase” on page 2:8* for the Chemical disinfection and cleaning procedure.

When the preservation procedure is completed the alarm *404 RINSE REQUIRED* appears in the display.

2.4 Conductivity monitoring

2.4.1 Rejection rate

There are two conductivity sensors measuring in the feed water line (C1) and product water line (C2). With those two values a percentage value can be calculated that indicates the salt rejection rate. The temperature sensor at the product water (T1) is used to compensate both conductivity measurements.

Calculating the rejection rate

The notification limit for the rejection rate is a preset value, refer to *“Rejection rate and Conductivity start” on page 2:36*, calculated by the formula below.

$$\text{Rejection rate} = \left(1 - \frac{\text{Conductivity in Product water line}}{\text{Conductivity in Feed water line}} \right) \times 100$$

2.4.2 Preset of conductivity limits

Settings of notification, alarm and stop limits can be changed in the internal service menu on the operator's panel or with the GXP software.

Presetable variables with descriptions and functions refer to *“Conductivity (Cond)” on page 2:35*.

Information about Conductivity notification, alarm and stop and Low rejection rate notification refer to *Operator's Manual*.

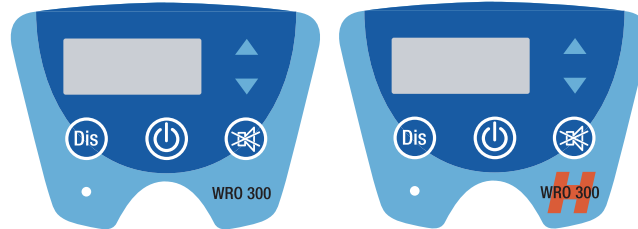
Guide of conductivity setting

The Conductivity limits should be set in relation to the feed water conductivity. Local regulation got to be considered when determining the values of the limits.

2.5 Main components

2.5.1 Operator's panel

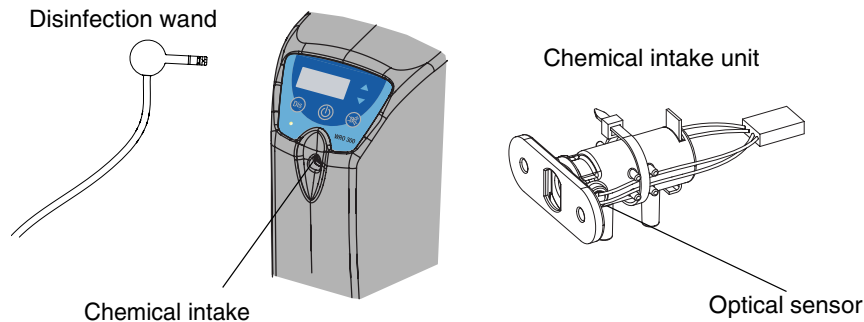
The circuit board with the display, buttons, buzzer and indication is covered with an overlay.



To check the operator's panel, refer to [“Testing the operator's panel” on page 6:7.](#)

2.5.2 Chemical intake unit

This unit supplies the WRO with chemicals for disinfection, cleaning or preservation. The optical sensor detects if the disinfection wand connector is inserted.

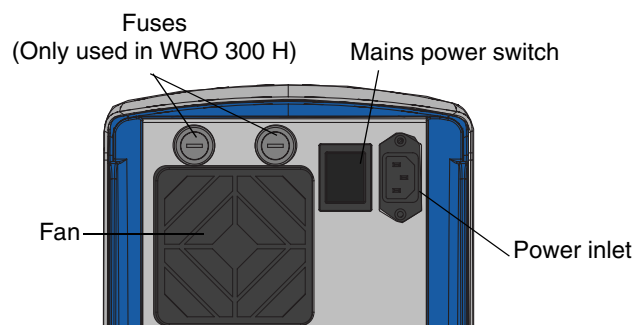


2.5.3 Power supply unit

The power supply can be 100, 115 or 220-240 V AC with 50 or 60 Hz and the Power supply unit adjusts automatically to the supplied voltage.

The internal voltage supply is 24 V DC. Except for the Heaters that are supplied with Main power voltage, refer to [“Heat” on page 2:37.](#)

The fan is cooling the power supply unit.



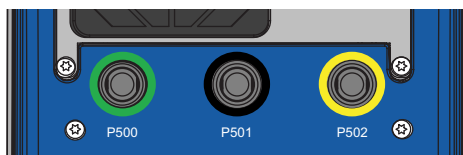
Fuses for WRO 300 H

Voltage	Fuse size	Gambro order no
100 - 130 V AC	15 A slow blow	W502 007 001
220 - 240 V AC	10 A slow blow	W502 006 002

2.5.4 External communication unit

External communication such as remote start from the dialysis machine, logging of data to a PC and downloading program from a PC is performed through these terminals.

At the rear side of the WRO there are three external communications terminals. The required cables are optional, ordered from Gambro according to the spare part numbers.



Port	Function	Description
P500 Green ^a	External communication Including remote Start/Stop	Communication between the WRO and newer versions of dialysis machines. ^b
P501 Black	Software communication	Connection to PC
P502 Yellow	Remote Start/Stop	Connection to remote control the WRO from the dialysis machine.
	External alarm	Can be connected to external indication. (i.e buzzer or lamp)

a. Blue in older version

b. NOTE Only recommended cables shall be used.

2.5.5 CPU board

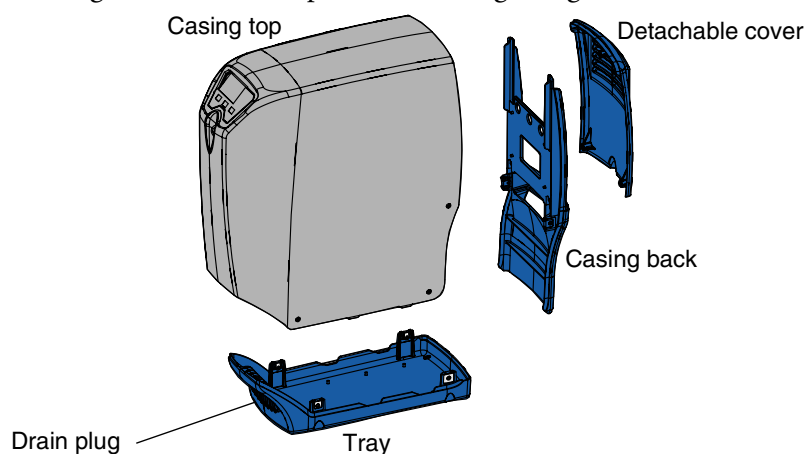
The CPU board contains most of the electronics that controls the function of the WRO.

2.5.6 AC Relay board (only WRO 300 H)

The AC relay board contains the electronics that controls the Heater rods.

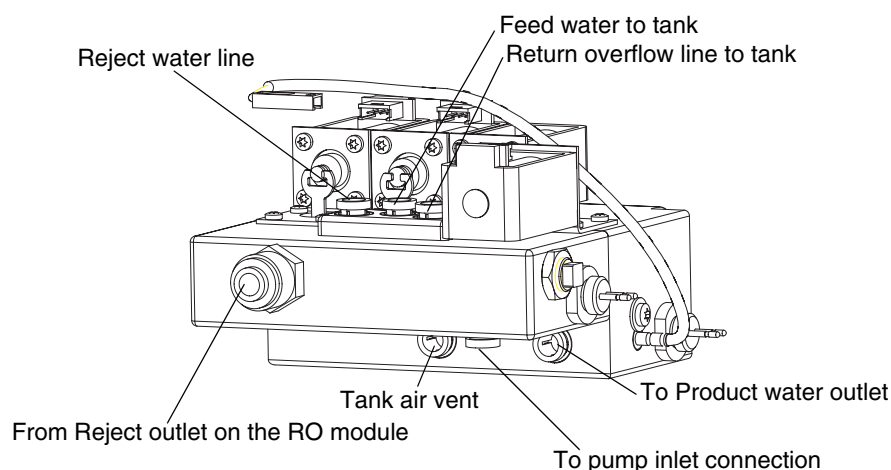
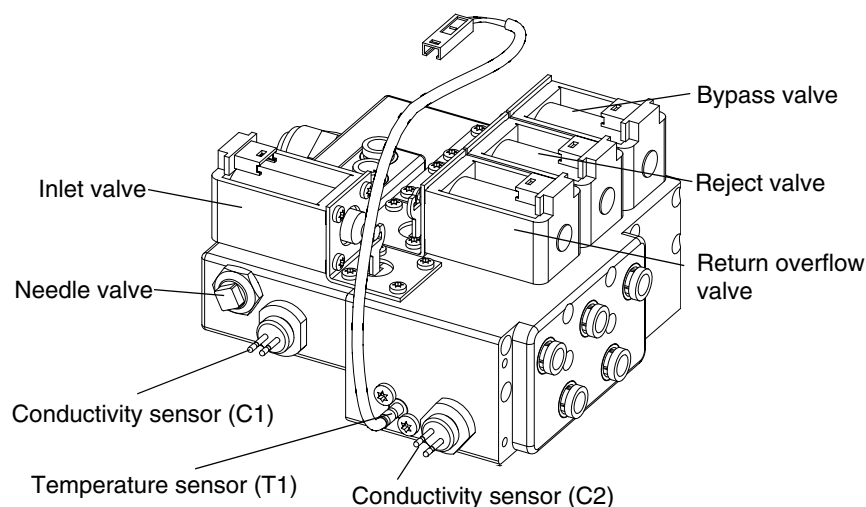
2.5.7 Casing

The casing consists of four pieces according to figure below.



2.5.8 Flow block

This block contains valves and sensors to control the flow in the WRO.



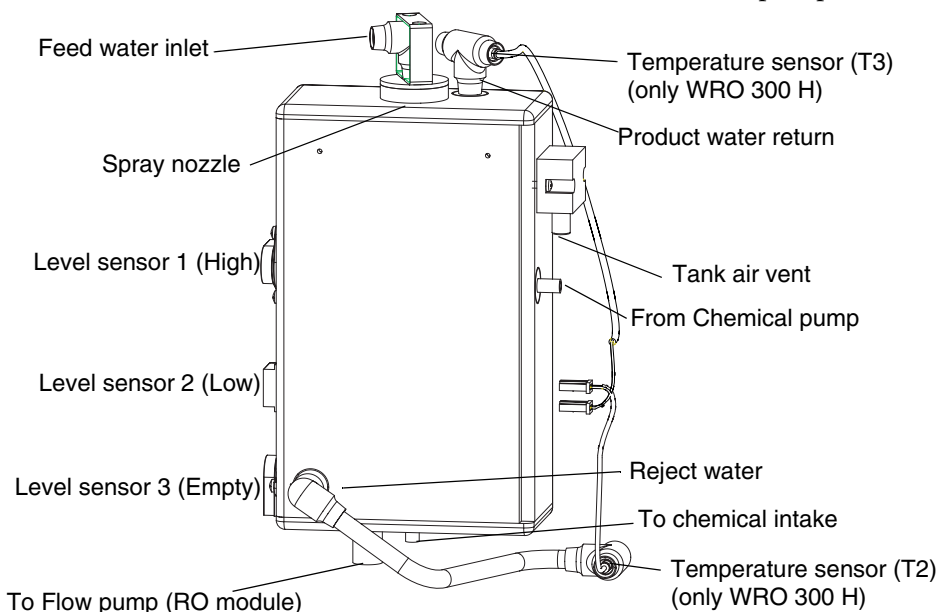
Component	Description
Reject valve (REVA)	The three way solenoid valve either divert to drain or tank.
Inlet valve (INVA)	Open and closes for feed water.
Bypass valve (BYVA)	Opening this valve the flow bypass the Flow valve.
Return overflow valve (OFVA)	This valve contains two functions with a solenoid valve function and a overflow valve function. When not actuated it creates a pressure to keep the flow rate of the product water to the dialysis machine. When actuated the pressure is released from the product water line.
Conductivity sensors	There are two conductivity sensors in the WRO. One is positioned in the feed water line (C1) and the second in the product water line (C2).
Temperature sensor	Measuring the temperature at the product water line (T1).
Needle valve	The needle valve is used to adjust the feed water pressure of the RO module to deliver the desired amount of product water.
Flow valve	A restrictor valve gives a steady flow of 1200 ml/min $\pm$ 100 ml/min through the Flow valve.

2.5.9 Y coupling and product water loop

The product water loop consists of two hoses from the WRO connected to the Y coupling. The hose from third connection at the Y coupling is connected to the dialysis machine.

2.5.10 Tank

The tank is supplied with water from the feed water inlet. Product return water from the Y coupling also enters the tank and if the Water saving function is in use reject water from the Reject valve returns to the tank. The water is distributed to the RO module or to the Chemical pump.



Spray nozzle

The feed water enters the tank unit through a spray nozzle.

Level sensors

There are three level sensors on the tank to indicate empty, low and high level.

The sensors may be cleaned with a gentle detergent or soap. No alcohol based substance may be used.

NOTE

Avoid to touch the sensors detection area with fingers or expose them to dust.

Tank air vent

The tank air vent outlet ensures an airgap in the tank.

2.5.11 RO module

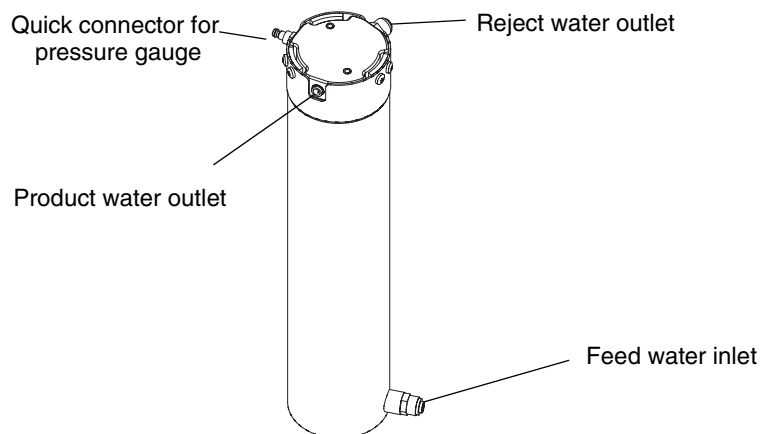


Fig.2.8 RO module for WRO 300 H.

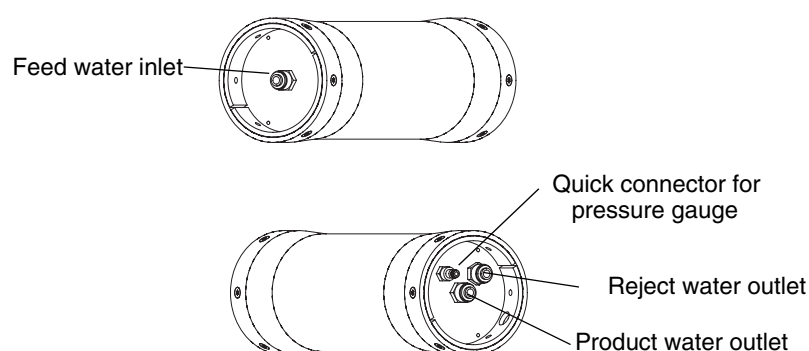


Fig.2.9 RO module for WRO 300.

The feed water is purified in the membrane using the reverse osmosis. The purified water goes to the product water outlet and the reject water to the reject water outlet.

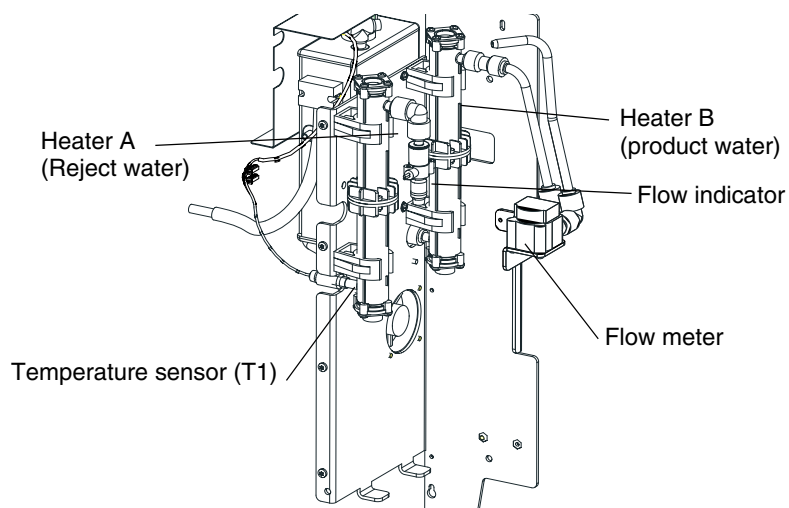
2.5.12 Heaters

The WRO 300 H is equipped with two Heaters with different properties. As the electrical connection are different it is not possible to connect the Heaters to the wrong connection.

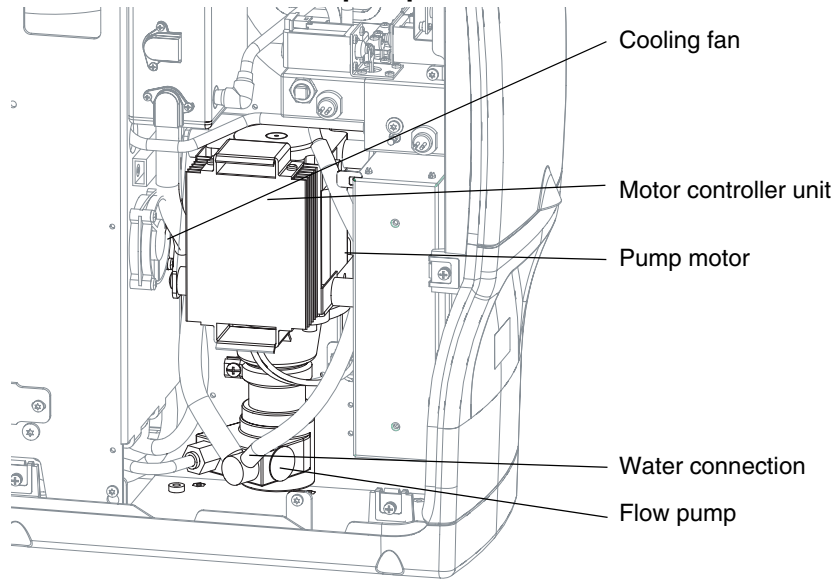


CAUTION

Although it is important to mount the Heaters in the right position.



2.5.13 Motor and Flow pump unit



The Flow pump circulate the water in the fluid path. It is driven by a 24 V DC brushless motor with a controller unit.

Cooling fan

To cool the pump and the motor unit a separate fan is mounted nearby.

2.6 Exchanging components

Follow the descriptions below to exchange components.

When disconnecting the tubes press the plastic ring carefully with a suitable tool, for example the silicon Connector Tool (Gambro order no. K12 755 001).

!

CAUTION

When exchanging components in the WRO the EMC plate has to be put back correctly. Otherwise the EMC/EMI protection is not in place.

!

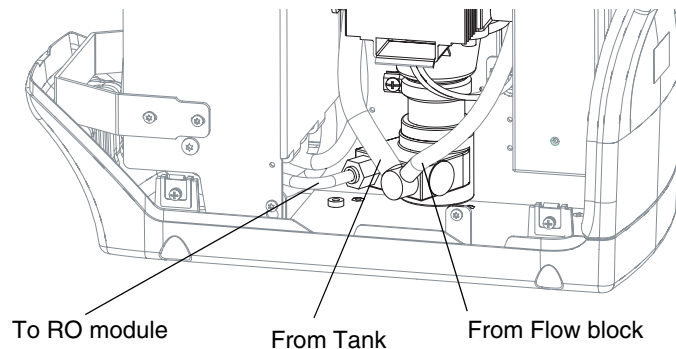
CAUTION

At delivery the WRO contains a preservation solution. In other modes different chemical solutions can be present. Take preventive cautions to avoid contact with those solutions.

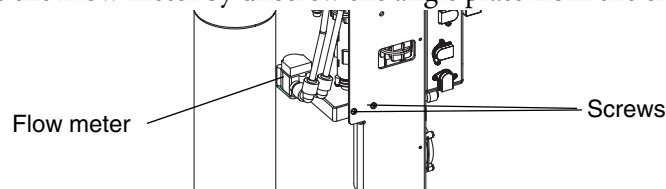
2.6.1 Exchanging RO module

Remove RO module, WRO 300 H

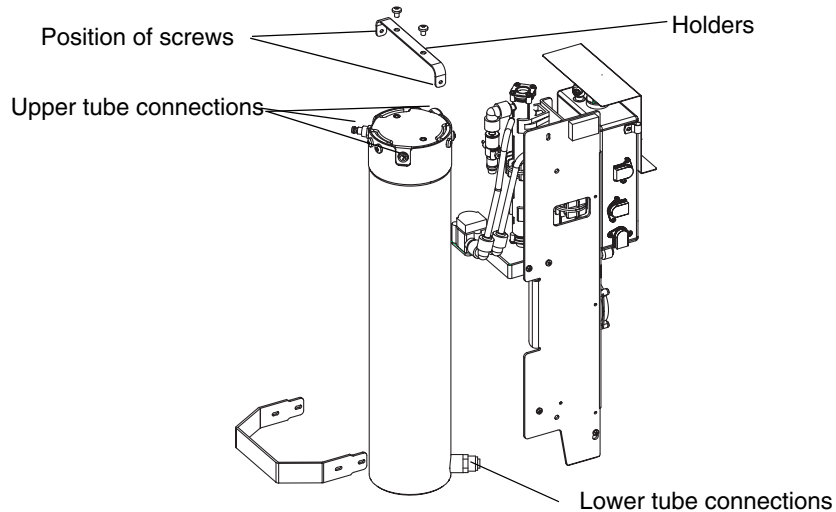
- 1 Stop the WRO 300 and turn off the power with mains power switch.
- 2 Remove the casing top.
- 3 Remove the EMC plate.
- 4 Remove the tubes from the Flow pump connections. Pour out the water into a container or drain.



- 5 Disconnect the lower tube at the Flow pump outlet connection and block the tube.
- 6 Remove the Flow meter by unscrew the angle plate from the chassi.



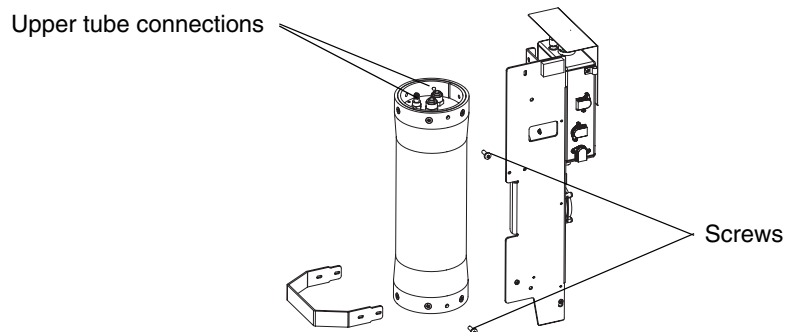
- 7 Remove the RO module by loosen the screws to the holder.



- 8 Remove the lower tube at the bottom of the RO module and pour out the water into a container or drain. Also remove the holder.

Remove RO module, WRO 300

- 1 Perform [step 1](#) to [step 5](#) in the previous instruction, refer to [“Remove RO module, WRO 300 H” on page 2:18](#).
- 2 Disconnect the two upper tubes.
- 3 Unscrew the two screws at the chassis and remove the RO module.
- 4 Remove the lower tube at the bottom of the RO module and pour out the water into a container or drain.



Mount the new RO module, WRO 300 and WRO 300 H



CAUTION

The RO module is filled with preservation and anti-freeze solution at delivery. Take preventive cautions to avoid contact with this solution.

- 1 Turn the RO module upside down to keep the preservation solution inside the membrane. Remove the plug before connecting the lower tube and block the open end.
- 2 Do the rest of the procedure above in reverse order when mounting the new RO module.
- 3 After the exchange of the RO module a rinse must be performed.
- 4 Connect the pressure gauge at the quick connector on the RO module.
- 5 Insert the disinfection wand connector into the chemical intake.
- 6 When **CHEM SELECT** is displayed press the down arrow button  to enter the list of available cleaning, disinfection and preservation programs.
- 7 Select preservation protocol.
- 8 Press  for three seconds.
- 9 Disconnect the disinfection wand connector. **404 RINSE REQUIRED** is displayed and press  for three seconds to initiate rinse.
- 10 Perform a residual test according to the procedure in *"Residual test after preservation" on page 3:3*.
- 11 Adjust the flow, refer to *"Adjusting the flow and the pressure" on page 3:4*.

2.6.2 Exchange Heaters

Remove Heaters

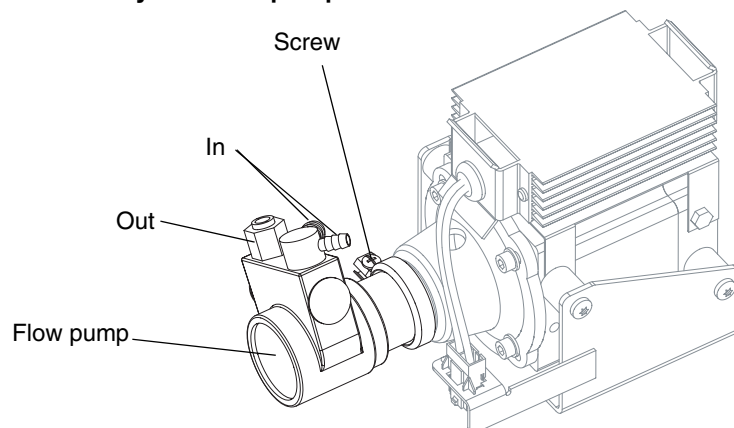
- 1 Remove the RO module according to description *"Remove RO module, WRO 300 H" on page 2:18*.
- 2 Disconnect the electrical connections at the AC relay board.
- 3 Disconnect upper and lower hydraulic connections at the Heaters.
- 4 Remove the Heaters by pulling them from their holders.

Mount the new Heaters

- 5 Do the procedure above in reverse order when mounting the new Heaters.

2.6.3 Exchanging Flow pump and motor

Remove only the Flow pump



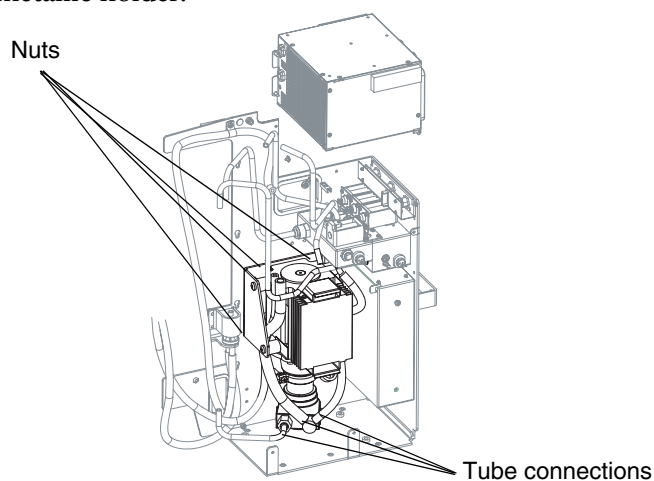
- 1 Stop the WRO and turn off the power with mains power switch.
- 2 Remove the casing top.
- 3 Remove the EMC plate.
- 4 Disconnect both inlet tubes (from the tank and the Flow block) and pour out the water from the tank into a container or drain.
- 5 Disconnect the outlet tubes and block it.
- 6 Unscrew the ring that holds the Flow pump attached to the Flow pump motor. Remove the Flow pump.

Mount the new pump

- 1 Put the new pump and ring back in the right position.
- 2 Connect the tubes and the perform [step 1](#) to [step 3](#) in the instruction above in reverse order.
- 3 Adjust the flow, refer to [“Adjusting the flow and the pressure” on page 3:4.](#)

Remove the Flow pump and motor unit

- 1 Stop the WRO and turn off the power with mains power switch.
- 2 Remove the casing top.
- 3 Remove the EMC plate.
- 4 Disconnect both inlet tubes (from the tank and the Flow block) and pour out the water from the tank into a container or drain.
- 5 Also disconnect the tube between the Chemical pump and the tank.
- 6 Disconnect the outlet tubes and block it.
- 7 Disconnect all cables connected to Flow pump controller unit.
- 8 Unscrew the four nuts. Then remove the Flow pump and motor unit.
- 9 Disconnect the electrical connector from the metallic holder. Then unscrew the four screw with rubber holders and remove the pump from metallic holder.



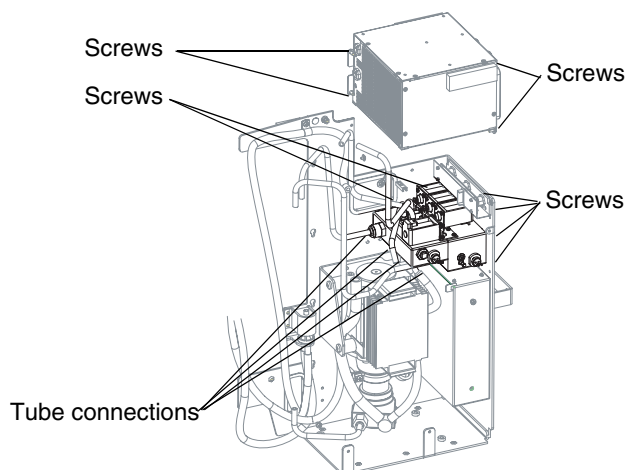
Mount the new Flow pump and motor unit

- 1 Do procedure above in reverse order, starting with *step 9* to *step 4*. It is necessary to adjust the needle valve before continuing with *step 3* and *step 1*.
- 2 Adjust the flow, refer to *“Adjusting the flow and the pressure” on page 3:4*.

2.6.4 Exchanging Flow block

Remove the Flow block

- 1 Stop the WRO and turn off the power with mains power switch.
- 2 Remove the casing top.
- 3 Remove the EMC plate.
- 4 Turn off the feed water supply.
- 5 Disconnect the external tubes at the rear side of the WRO.
- 6 Remove the casing back by unscrewing the four screws.
- 7 Unscrew the power supply unit. Turn it upside down and lay it on the top of the RO module.
- 8 Disconnect the cables to the two conductivity sensors, the temperature sensor and the four solenoid valves.
- 9 Disconnect all internal tube connections.
- 10 Unscrew the four screws at the rear side of the WRO and two at the side of the chassis.



Mount the new Flow block

- 1 Do the procedure above in reverse order, starting with [step 10](#) to [step 4](#).
- 2 Check for leakage.
- 3 Adjust the flow, refer to *“Adjusting the flow and the pressure” on page 3:4*.
- 4 Check the conductivity reading, refer to *“Conductivity monitoring” on page 2:11*.
- 5 Finish with [step 3](#) to [step 1](#).

2.6.5 Exchanging Power supply unit

The table below show which version of Power supply units can be used.



CAUTION: DISCONNECT POWER BEFORE CHANGING FUSE. REPLACE ONLY WITH SAME TYPE AND SAME RATING OF FUSE.

ATTENTION: COUPER LE COURANT AVANT DE REMPLACER LE FUSIBLE. REMPLACER PAR UN FUSIBLE DE MEME TYPE ET DE MEMES CARACTERISTIQUES NOMINALES.

警告：更换保险丝之前请断开电源连接！

		F1	F2
WRO 300	100 - 240 V AC		
WRO 300 H	220 - 240 V AC	10 A	10 A
WRO 300 H	100 - 115 V AC	15 A	15 A



W20045006

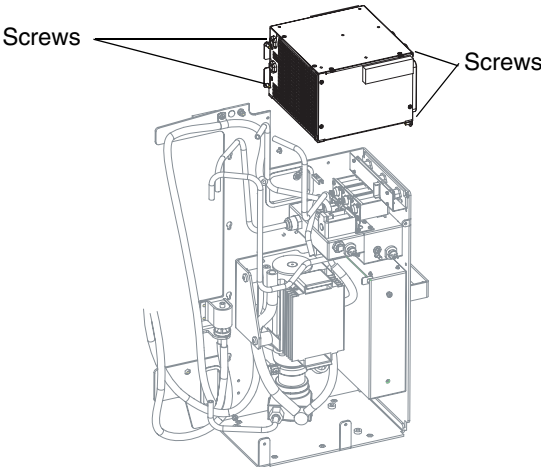
Part no.

Contact Gambro to get order no.

W200 045 xxx		001	002	003	004	005	006
WRO 300	100 - 130 V AC					X	X
	220 - 240 V AC	X	X	X	X	X	X
WRO 300 H	100 - 130 V AC						X
	220 - 240 V AC	X			X		X

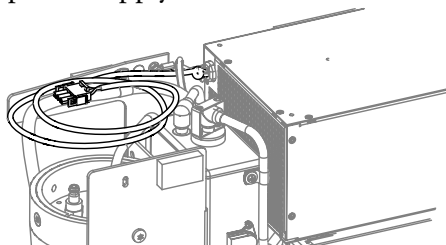
Remove the power supply unit

- 1 Stop the WRO and turn off the power with mains power switch.
- 2 Disconnect the mains power cable.
- 3 Remove the casing top.
- 4 Disconnect the protective earth cable.
- 5 Remove the EMC plate.
- 6 Disconnect the cables from the pump unit and the CPU board. In the WRO 300 H also disconnect from the AC Relay board.
- 7 Unscrew the four screws and remove the power supply unit



Mount the new power supply unit

- 1 Before mounting the new power supply unit check the fuses.
In WRO 300 remove the fuses.
In WRO 300 H insert fuses according to Mains voltage, refer to *"Fuses for WRO 300 H" on page 2:12.*
- 2 Do the procedure above in reverse order when mounting the new power supply unit.



!

CAUTION

Check that the protected earth is correctly attached to the chassis.

!

CAUTION

In WRO 300 H make sure that the protective earth is correctly attached to the Heaters.

2.6.6 Exchanging CPU board, WRO 300



CAUTION

When exchanging the CPU board ESD protection is required.

Remove the CPU board

- 1 Stop the WRO 300 and turn off the power with mains power switch. Disconnect the mains power cable.
- 2 Remove the casing top.
- 3 Disconnect all cables from the CPU board.

NOTE

Pull the cable connectors straight out from board to avoid damage on the board mounted connectors.

- 4 Remove the metal holders.
- 5 Pull out the CPU board gently.

Mount the new CPU board

- 1 Do the procedure above in reverse order when mounting the new CPU board.

NOTE

When mounting the new CPU board make sure that the board is correctly placed in the guides.

Also verify that all preset value and conductivity limits is set to the same settings as the old CPU board.

2.6.7 Exchanging CPU board or AC relay board, WRO 300 H



CAUTION

When exchanging the CPU board or the AC relay board ESD protection is required.

Remove the CPU board and AC relay board

- 1 Stop the WRO 300 H and turn off the power with mains power switch. Disconnect the mains power cable.
- 2 Remove the casing top.
- 3 Disconnect all cables from the CPU board and AC relay board.

NOTE

Pull the cable connectors straight out from board to avoid damage on the board mounted connectors.

- 4 Remove the metal holder.
- 5 Pull out the CPU board and AC relay board simultaneously.
- 6 Then disconnect the short cable between the boards.
- 7 Mount the new CPU board and AC relay board
- 8 Do the procedure above in reverse order when mounting the new CPU board and AC relay board.

NOTE

When mounting the new CPU board and AC relay board make sure that the board is correctly placed in the guides.

Also verify that all preset value and conductivity limits is set to the same settings as the old CPU board.

2.7 Internal Service

Change value with up or down arrow.

Press  (OK) to continue or confirm.

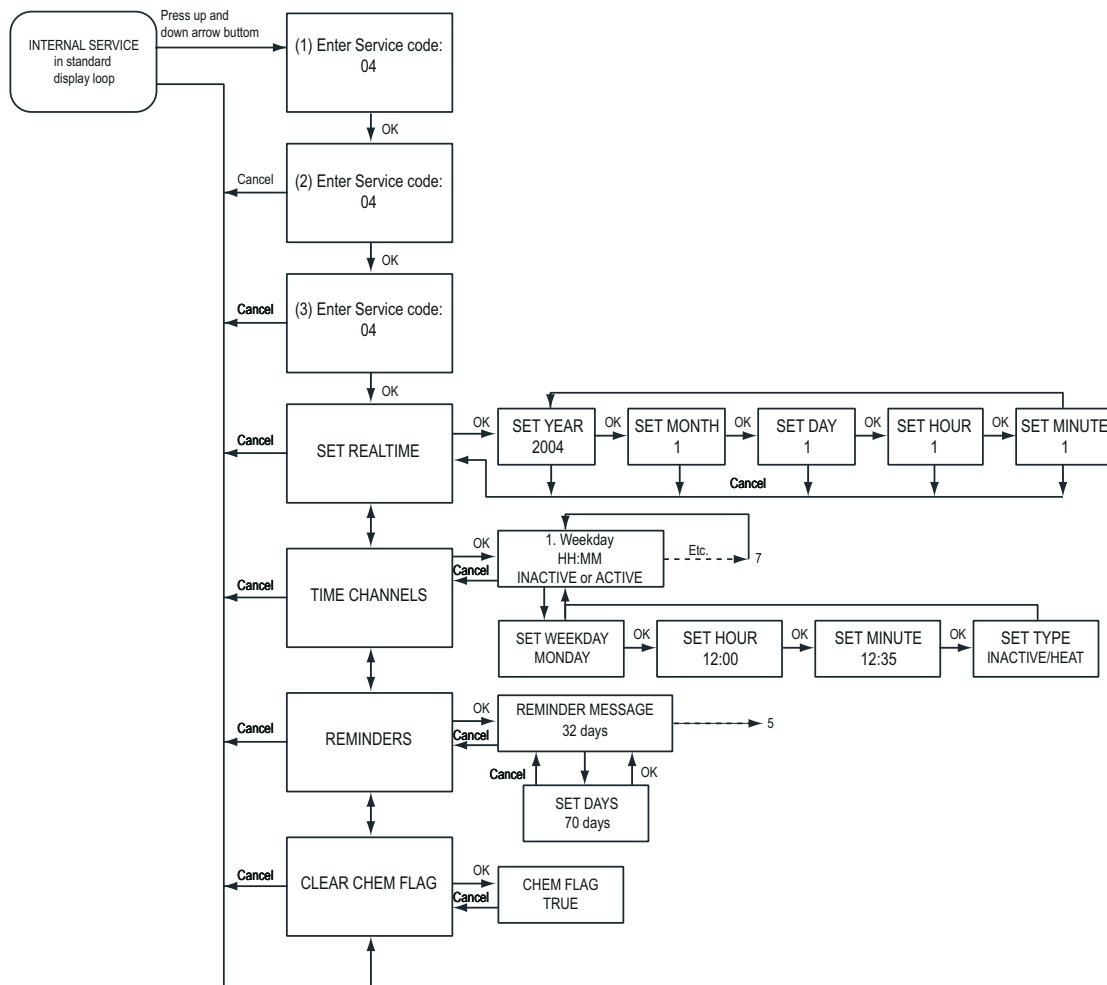
Press  (cancel) to cancel and leave menu.

Service code	16 90 00
--------------	----------

2.7.1 To enter internal service

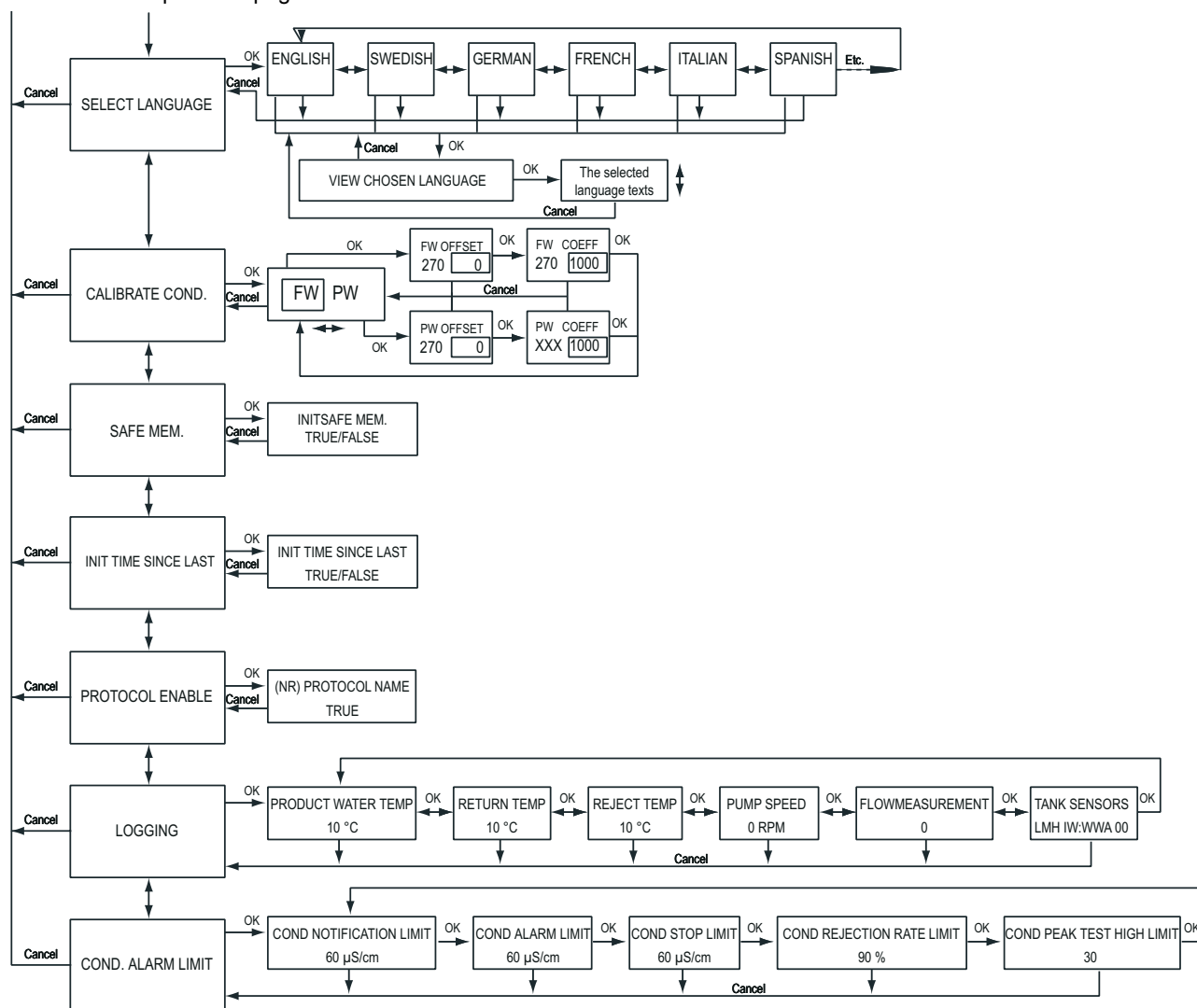
- 1 Browse using the arrow buttons the standard loop until "INTERNAL SERVICE".
- 2 Press both arrow buttons at the same time to enter to the menu for service code 1 (16) use the arrow buttons to change the value, and then press  (OK).
- 3 Enter the service code 2 (90) using the arrow buttons, and then press  (OK).
- 4 Enter the service code 3 (00) using the arrow buttons, and then press  (OK).

See below for more details.



Continue on next page

Continues from previous page



Menus	Description
ENTER SERVICE CODE	Enter service code to access Internal Service mode. The six number codes are entered in three steps with two numbers in each step.
SET REALTIME	A menu with the functionality to set date and time
TIME CHANNELS	A menu with the functionality to set up seven different timers, used to start selected function.
REMINDERS	A menu with the functionality to change the set up time, for five different reminders.
CLEAR CHEM. FLAG	A menu with the functionality to change the "CHEM. FLAG" status.
SELECT LANGUAGE	A menu with the functionality to select the menu language, and a possibility to scroll the menu texts.
CALIBRATE COND.	A menu with the functionality to calibrate the feed water (C1) and product water (C2) conductivity sensor
SAFE MEM.	A menu with the functionality to reset the safe memory area to default values, see below. lastAcid = 0 lastAlk = 0 lastHeat = 0 lastChem = 0 inletValveCount = 0 bypassValveCount = 0 returnValveCount = 0 rejectValveCount = 0 chemPump = 0 roPump = 0 visualStateMode = E_COLD_START condPulsStopTime = 0 lastUsedChemProt = 0 chemFlag = 1
INIT TIME SINCE LAST	A menu with the functionality to reset all the "TIME SINCE LAST" timers to zero.
PROTOCOL ENABLED	A menu with the functionality to change the chem. protocol visibility for the CHEM SELECT menu.
LOGGING	A menu to view selected values, such as different temperature values, pump speed, flow measurement, etc.
COND. ALARM LIMIT	A menu with the functionality to change the different conductivity alarm limits.

2.8 PC support software

2.8.1 GXL - Logging

Logging of parameters like conductivity, flow, temp etc.

For detailed information of the GXL software refer to the *Operator's Manual* of the software.

2.8.2 GXP - Preset

The service technician can change all presets by the use of the Gambro service tool, GXP. The GXP runs on an ordinary PC communicating with the WRO through a RS-232 interface. The technician starts the GXP session by first putting the WRO in Standby mode. In the GXP software enter 16 90 00 as access code. The service technician must switch off and on the WRO in order enter Standby mode again. This will force the WRO to restart the PC; therefore will all preset parameters be re-read before use of the WRO.

For detailed information of the GXP software refer to the *Operator's Manual* of the software.

2.8.3 GWD - Download

New program versions are downloaded to the flash memory in the WRO from the PC.

Upgrade the program using a PC

The WRO shall already have at least program version 2.0 or later.

Start the GWD

- 1 Double click on the GWD.exe to start the program.

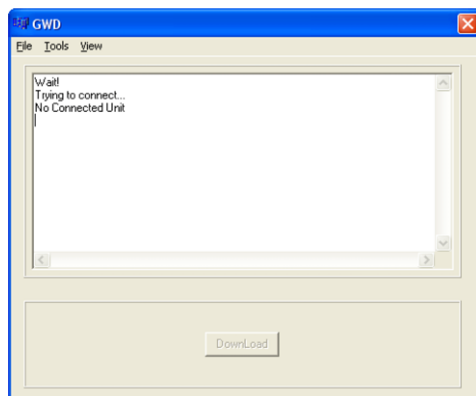
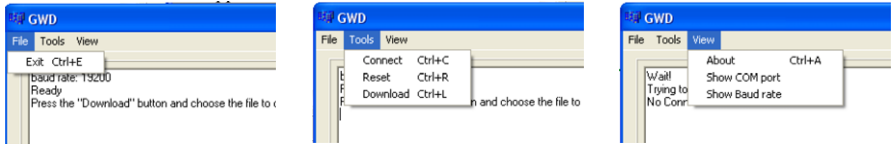


Fig.2.10 Main window.

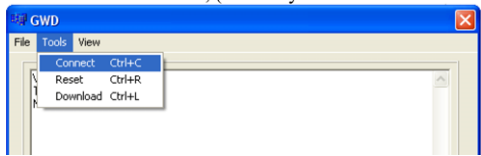
Menus



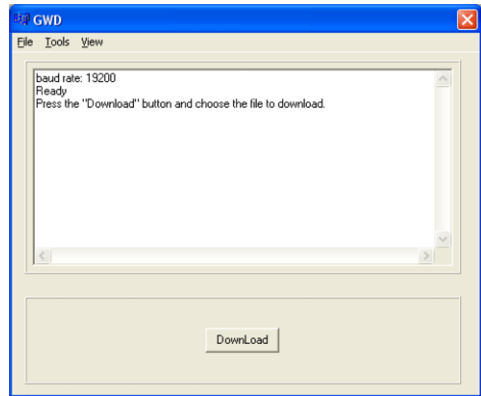
Menus	Description	
File	Exit	Ends the GWD program.
Tools	Connect	Connects to a WRO, if the WRO is put in to Boot Loader position, normally the program makes an auto connection when start up.
	Reset	Clears the data like the COM port and Baud rate etc.
	Download	Activates the download function.
View	About	Displays data about the program
	Show COM port	Displays which COM port the WRO is connected to.
	Show Baud rate	Displays the the Baud rate

Download a new program

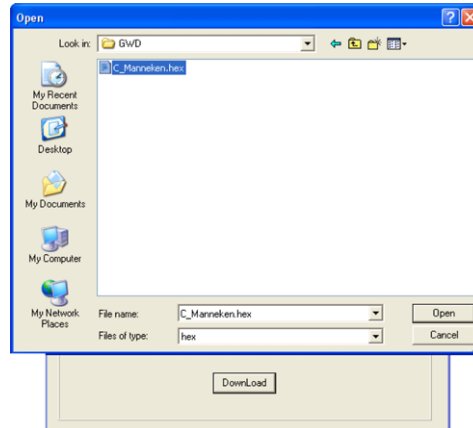
- 1 Connect a communication cable to one of the COM ports
- 2 Start the WRO, and at the same time press the  button, hold it down until the LCD turns black.
- 3 Select *Connect* in the menu *Tools*, and the program will set up a connection to the WRO. Normally the connection is established automatically, if not try to do a manually start, refer to *“Manual connection” on page 2:34.*



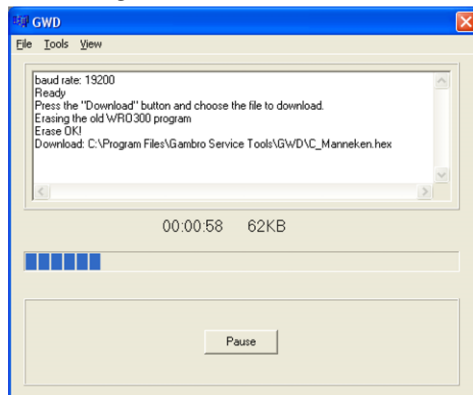
- 4 Click on *Download*.



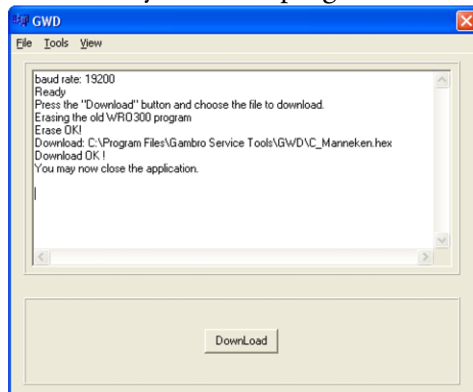
- 5 Select the file to download.



- 6 The program will proceed the download, displaying the status in the dialog box.



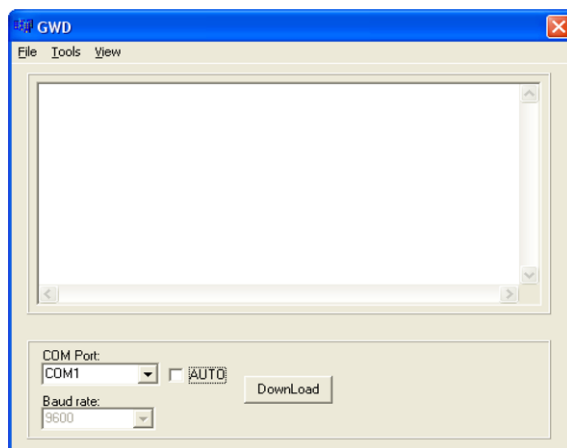
- 7 When ready close the program.



- 8 The GWD verifies that the download was performed correctly. When the download is confirmed as OK, restart the WRO using the power switch.

Manual connection

- 1 If connection is not established to the WRO, restart the WRO in Boot loader position.
- 2 Select *Reset* in the menu *Tools*.
- 3 Select *Show Baud* rate in the menu *View* and uncheck the *AUTO* box.
- 4 Select COM port for communication and set the Baud rate to 9600 baud.



- 5 Select *Connect* in the menu *Tools*.
- 6 Then continue to download the program, refer to [“Download a new program” on page 2:32](#).

2.9 Presets and defaults

The service technician can change presets by the use of the Gambro service tool, GXP or via the Internal Service mode. Refer to [“GXP - Preset” on page 2:31](#) or [“Internal Service” on page 2:28](#).

The presets are grouped in order to support the technician when using the GXP software. The presets are divided into the following groups.

2.9.1 Flow pump

Parameter	Settings
Name	CRP_RUN_SPEED_PRESET
S.no	S28
Preset group	Flow pump
Description	Output duty cycle for Flow pump during (RUN) operation.
Scale	1 %
Default	35
Min	0
Max	46

2.9.2 Water save

Parameter	Settings
Name	CVC_WATER_SAVE_FACTOR
S.no	S29
Preset group	Water save
Description	Set value for the time the reject water is redirected back to the tank in order to reduce water consumption.
Scale	1 %
Default	5
Min	0
Max	80

2.9.3 Conductivity (Cond)

The conductivity monitoring is using the three limits, notification, alarm and stop. For detail descriptions on how the conductivity monitoring see *Operator's manual*.

Alarm limits

Parameter	Settings		
Name	CCM_NOTIFICATION	CCM_ALARM	CCM_STOP
S.no	S33	S34	S35
Preset group	Cond	Cond	Cond
Description	Notification conductivity limit. Conductivity notification alerts the user when entering Standby	Conductivity alarm limit	Conductivity stop limit. WRO stops when this limit is exceeded
Scale	1 µS/cm	1 µS/cm	1 µS/cm
Default	60	60	60
Min	0	0	0
Max	500	500	500

Rejection rate and Conductivity start

Parameter	Settings	
Name	CCM_REJECTION_RATE_LIMIT	CCM_PEAK_TEST_HIGH_LIM
S.no	S149	S152
Preset group	Cond	Cond
Description	Rejection ratio notification limit	Enable or disable Peak test
Scale	1 %	>6 = Peak test enabled =<6 = Peak test disabled
Default	90	30
Min	0	0
Max	98	200

2.9.4 User interface panel (LCD)

Parameter	Settings		
Name	LCD_PRES_LANGUAG E	CLC_BACKLIGHT_TIME OUT	LCD_LOOP_TIMEOUT
S.no	S36	S143	S144
Preset group	LCD	LCD	LCD
Description	Chosen language for the LCD texts.	Chosen LCD backlight time out	Chosen LCD loop time out
Scale	0 = English 1 = Swedish 2 = German 3 = French 4 = Italian 5 = Spanish 6 = Danish 7 = Finnish 8 = Nederlands 9 = Portuguese 10 = Turkish 11 = Norweigian 12 = Czech	0 = For ever 1 - 60 = seconds	1 - 60 = seconds
Default	English (0)	15	15
Min	0	-1	1
Max	12	60	60

2.9.5 Heat

The Heaters consists of a number of windings and to obtain correct power a specific combination of those has to be activated. Different combinations are required depending on which power supply voltage that is used.

Heat supply voltage

Parameter	Settings
Name	CHC_SUPPLY_VOLTAGE
S.no	S32
Preset group	Heating
Description	Mains supply voltage for WRO Heater function
Scale	0 = non Heater function available 1 = 220 - 240 V AC 2 = 115 V AC 3 = 110 V AC
Default	non (0) ^a
Min	0
Max	3

a. The setting for WRO 300 H is changed before delivery to 230 V AC (1)

Heat Set temperature

Parameter	Settings	
Name	CHC_PRODUCT_HEAT_SETVAL	CHC_REJECT_HEAT_SETVAL
S.no	S30	S31
Preset group	Heating	Heating
Description	Set temperature of Heater B	Set temperature of Heater A
Scale	0.01 °C	0.01 °C
Default	9200	9200
Min	0	0
Max	9500	9500

Active cooldown

Parameter	Settings
Name	CHC_ACTIVE_COOLDOWN
S.no	S142
Preset group	Heating
Description	Enable Active cooldown
Scale	0 = disable 1 = enable
Default	1
Min	0
Max	1

Low flow heat

Parameter	Settings	
Name	CHC_LOW_FLOW_HEAT_SETVAL	CHC_LFH_AUTO_HEAT
S.no	S181	S183
Preset group	Heating	Heating
Description	Set temperature of Heater B in Low flow heat mode	If set to TRUE a Heat disinfection after a remote started Low flow heat
Scale	0.01 °C	0 = disable 1 = enable
Default	8300	1
Min	0	0
Max	8500	1

2.9.6 Disinfection protocol manager

The WRO offers the user to preset up to ten different disinfection/cleaning protocols.

All parameters included in the setup for each protocol are changeable through the GXP program. The Internal Service mode can be used to change the parameter indicating if a particular protocol shall be visible (selectable) during the chemical select procedure

Select protocol

The user selected which protocol to be used during the chemical selection procedure (Refer to *Operator's manual*).

Parameter	Settings
Name	CDM_PROT_PRESET
S.no	S138
Preset group	DisProtManager
Description	First protocol displayed in chemical select mode
Scale	Protocol number 1 to 10.
Default	1
Min	1
Max	10

Disinfection protocol

	Scale	Description	Min	Max
Visibility of protocol	1 = Selectable 0 = invisible	Indicates if disinfection protocol one shall be selectable or not.	0	1
Type of protocol	0 = Chemical disinfection 1 = Central chemical disinfection 2 = Preservation + Freeze 3 = Acid cleaning 4 = Alkali cleaning 5 = Membrane flush 6 = Preservation	Type of protocol	0	6
Protocol name	Text string	Protocol name, will be showed on the LCD during chemical select.	-	-
Dwell period	1 minute	Dwell period for the protocol.	0	3000
Intake volume	1 ml ^a	Volume to be used by the WRO during the intake phase of a disinfectant or cleaning cycle.	0	2500
Rinse time	1 minute	Rinse time	6	1000
Intermittent circulation during dwell period	1 = TRUE 0 = FALSE	Indicates if the WRO shall intermittent circulate the fluids during the dwell period.	0	1
Intermittent run time	1 minute	The intermittent run time, used when "Intermittent circulation during dwell period" true.	0	1000
Intermittent stop time	1 minute	The intermittent stop time, used when "Intermittent circulation during dwell period" is set to true.	0	1000
Second rinse time	1 minute	A second presetable rinse time, currently not used by the program.	0	1000

a. If Central Chem mode the setting scale is in minutes.

Default setting of disinfection protocols

Parameter name	Protocol									
	1	2	3	4	5	6	7	8	9	10
	S37-S46	S47-S56	S57-S66	S67-S76	S77-S86	S87-S96	S97-S106	S107-S116	S117-S126	S127-S136
Visibility of protocol	1	1	0	0	0	0	1	0	0	0
Type of protocol	0	3	2	1	1	1	6	0	3	4
Protocol name	DIALOX	ACIDCLEAN	PRESERV+FREEZE	CENTR_CH_95	CENTR_CH_200	CENTR_CH_200U	PRESERVATION	MINNOCARE.	MINNOCLEAN	ALKALICLEAN
Dwell period	30	30	0	30	30	30	0	30	30	30
Intake volume	115	250	0	12	13	18	500	50	200	250
Rinse time	60	30	60	60	60	60	60	60	30	30
Intermittent circulation during dwell period	0	1	0	0	0	0	0	0	1	1
Intermittent run time	2	5	0	0	0	0	0	0	5	5
Intermittent stop time	10	5	0	0	0	0	0	0	5	5
Second rinse time	30	30	30	60	30	30	30	30	30	30

Concentration of disinfectant

To calculate the concentration in the WRO following value is required.

Value	Description	Unit
V _m	Internal water volume in the WRO (3500 ml) Product water loop 2 x 1 m (100 ml)	ml
K _m	The wanted concentration of the solution in the WRO	ml
K _c	The concentration of the disinfectant	%
V _c	Intake volume of the disinfectant	%

Formula

To calculate	Formula
Concentration of the solution	$K_m = K_c * V_c / V_m$
Volume of disinfectant	$V_c = K_m * V_m / K_c$

Example: Intake volume of Dialox

- $V_m = 3500 + 100 \text{ ml}$
- $K_m = 3 \%$
- $K_c = 100 \%$

$$V_c = 3 \% * 3600 / 100 \%$$

$$V_c = 108 \text{ ml}$$

Example: Intake volume of prediluted chemical

- $V_m = 3600 \text{ ml}$
- $K_m = 3 \%$
- $K_c = 75 \%$

$$V_c = 3 \% * 3600 / 75 \%$$

$$V_c = 144 \text{ ml}$$

2.9.7 Time manager

Auto flush

Parameter	Settings	
Name	CTM_AUTOFLUSH_STANDBY_TIME	CTM_AUTOFLUSH_RUN_TIME
S.no	S139	S140
Preset group	Time manager	Time manager
Description	Time before an Auto flush starts	Time for an Auto flush period
Scale	The function is disabled if the value is equal to zero.	1 minutes
Default	0	1
Min	0	1
Max	10000	100

Time since last

Parameter	Settings			
Name	CTM_VISIBLE_TIME_SINCE_LAST_HEAT	CTM_VISIBLE_TIME_SINCE_LAST_CHEM	CTM_VISIBLE_TIME_SINCE_LAST_ACID	CTM_VISIBLE_TIME_SINCE_LAST_ALK
S.no	S153	S154	S155	S156
Preset group	Time manager	Time manager	Time manager	Time manager
Description	When enabled display shows time since last Heat disinfection	When enabled display shows time since last Chemical disinfection	When enabled display shows time since last Acid cleaning	When enabled display shows time since last Alkaline cleaning
Scale	0 = disable 1 = enable	0 = disable 1 = enable	0 = disable 1 = enable	0 = disable 1 = enable
Default	0	1	1	1
Min	0	0	0	0
Max	1	1	1	1

2.9.8 Time channels

Parameter	Settings
Name	CTS_TIME_CHANNEL_1 - CTS_TIME_CHANNEL_7
S.no	S159 - S165
Preset group	Time channel
Description	Set time for start of automatically Heat disinfection
Scale	hh:mmwt h = hour m = minute w = day of week (0 = Monday, 1 = Tuesday,...) t = Type (0 = disabled, 1 = Heat disinfection)
Default	00:0010 (text string)

2.9.9 Reminders

Parameter	Settings
Name	CRS_REMINDER_M_1 - CRS_REMINDER_M_5
S.no	S171 - S180
Preset group	Reminders
Description	Set text to each Reminder. 14 characters is available in each text string. By using the second preset variable (CRS_REMINDER_M_12 for reminder 1) another 14 character can be added.
Scale	Text string

2.9.10 Remote

Parameter	Settings
Name	CRC_WRO_AK_SERIAL_COM
S.no	S182
Preset group	Remote
Description	Setting of serial or parallel communication
Scale	0 = parallel 1 = serial
Default	0
Min	0
Max	1

2.9.11 Chemical intake pump

Parameter	Settings
Name	CCP_PUMP_CONSTANT_WATER
S.no	S151
Preset group	Chem
Description	Chemical intake pump calibration value. The value represents the pump capacity in terms of volume per seconds.
Scale	1/10 ml/sec
Default	18
Min	10
Max	30

3 Maintenance

3.1 Water testing

Depending on system setup and local conditions and regulations certain tests should be performed on a regular basis to insure proper function. The water sample should be collected at the inlet to the WRO. Following tests are recommended.

3.1.1 Total chlorine (chlorine + chloramines)

Perform a test for total chlorine. Exchange the carbon filter if the levels exceeds 0,1 mg/l.

Example of test kit

Stericheck Chlorine Sensitive, Gambro order no 103429.

3.1.2 Hardness (if a softener is installed)

Before each treatment

Test the water hardness according to the softener's manual. If the water is hard, check the salt level in the softener and, if required, add salt. The WRO can then be started

NOTE

WRO will not be damaged if operated with hard water for shorter periods of time (one week).

Example of test kit

Stericheck Hardness, Gambro order no 103430.

3.2 Changing the prefilter cartridge(s)

Frequency

Monthly or according to local recommendations.

Example of cartridge

Carbon block filter (10 inch), Gambro order no 24-10101.

3.2.1 Procedure

NOTE

The procedure below is valid for a Gambro carbon block prefilter.

- 1 Turn off the water supply.
- 2 Relieve the pressure in the prefilter housing. This is normally done by depressing the red button on the top of the prefilter housing.
- 3 Unscrew the prefilter housing and remove the old cartridge.
- 4 Clean the inner surface of the housing.
- 5 Insert a new cartridge and reinstall the housing.
- 6 Turn on the water supply. Check for possible leakage.
- 7 Vent the prefilter by pressing the red button until all air is expelled.

3.3 Preservation with glycerol

!

CAUTION

Avoid eye contact and prolonged skin contact when using formaldehyde. Frequent and prolonged contact may cause skin irritation. If contact with skin or eyes occurs, flush the exposed area with water. Contact a physician if irritation develops.

If freeze protection is required when preserving the WRO is has to be performed with glycerol. This preservation solution is entered through the Feed water inlet and an external pump is required.

Procedure

- 1 Stop the WRO.
- 2 Disconnect the Feed water tube and connect the tube from container with the preservation solution via the external pump.
- 3 Insert the disinfection wand to the Chemical intake.
- 4 **CHEM SELECT** is shown in the display and is **Dis** flashing.



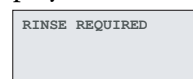
- 5 Choose a chemical protocol suitable for the preservation with freeze protection.
- 6 Press **Dis** for three seconds to initiate the intake.



- 7 When the intake is finished **501 REMOVE WAND** is displayed.

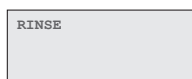


- 8 After approximately five minutes circulation of the fluid the preservation procedure is completed when **RINSE REQUIRED** is displayed.



3.3.1 Rinse

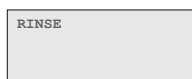
- 1 The indication  and  flash and the display shows *RINSE*



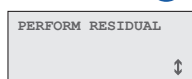
NOTE

If  is pressed the display will show *RINSE REQUIRED*. Steady light on indication .

- 2 Press  for three seconds to initiate rinse. (Steady light of the indication ).



- 3 When the rinse program is completed the display will show *PERFORM RESIDUAL TEST*. The indication  will remain steady lit and the  will flash.



3.3.2 Residual test after preservation

Formaldehyde

- 1 Press  for three seconds to start the WRO. Indication  will be steady lit and  will flash.
- 2 Collect a water sample from the product water line.
- 3 Verify that no residual disinfectant can be detected in the sample.
- 4 Remove any tag indicating storage with disinfectant.

Memstor

The required rinse program in the WRO has been designed and validated to insure complete rinse-out of the preservation and anti-freeze solution.

- 1 Press  for three seconds to start the WRO. Indication  will be steady lit and  will flash.
- 2 Let the WRO run in normal operation for at least 5 minutes. Then press  or  to show the conductivity value in the display (Within 30 seconds the display will return to *PERFORM RESIDUAL TEST*). check and verify that the product water conductivity remains stable and constant.
- 3 Confirm by pressing  until light goes out (three seconds). Indication  remains steady lit.



- 4 The WRO is now ready to use.

3.3.3 In order to interrupt procedure

Pause of intake phase

To pause the chemical intake press **Dis** for three seconds. To continue the intake press **Dis** for three seconds again.

Interrupt intake phase

To interrupt the chemical intake disconnect the disinfection wand. The intake phase restarts when the disinfection wand is connected again.

Interrupt dwell period phase

By pressing **Dis** for three seconds the disinfection procedure is interrupted. **501 REMOVE WAND** appears in the display and when the wand is disconnected the rinse procedure starts.

3.4 Component maintenance

3.4.1 Power supply unit

Fan filter

A fan filter is required to protect the power supply unit from dust. Depending on the environment the filter has to be exchanged regularly to get the right air flow through the unit.

Protective Earth (PE)

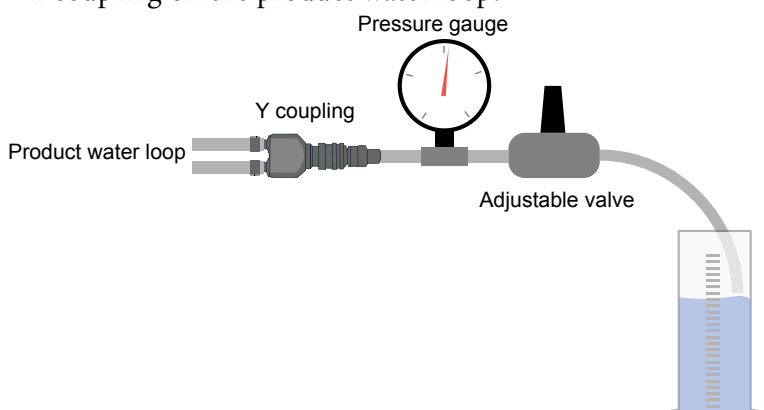
Verify the Protective Earth connection at regular intervals.

3.4.2 RO module

To keep the membrane in good condition it needs regular cleaning and disinfection. Refer to the *Operator's Manual*.

3.5 Adjusting the flow and the pressure

- 1 Mount a pressure gauge in the quick connector on the RO module.
- 2 Mount another pressure gauge and an adjustable valve after the Y coupling on the product water loop.



- 3 Start the WRO and let it run for 30 minutes.
- 4 Adjust the valve until the pressure in the product water loop is 1 - 2 bar.
- 5 Measure the product water flow. Adjust the needle valve in the Flow block until the flow is 1200 ml/min.
- 6 Check the reject pressure at the pressure gauge on the RO module. The pressure should be below 13 bar.

4 Wiring diagram

4.1 WRO 300 H

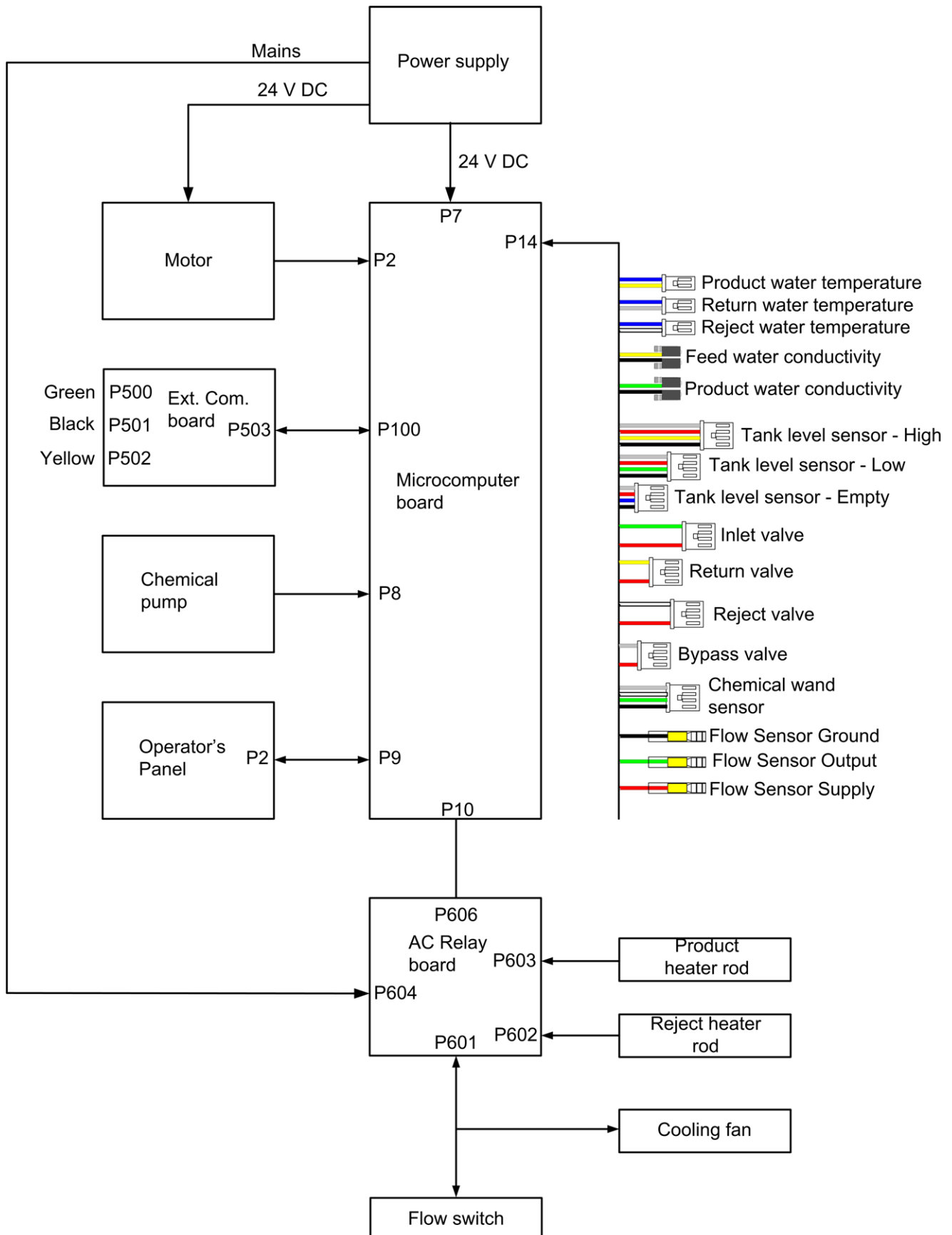


Fig.4.1 Wiring diagram for WRO 300 H

4.2 WRO 300

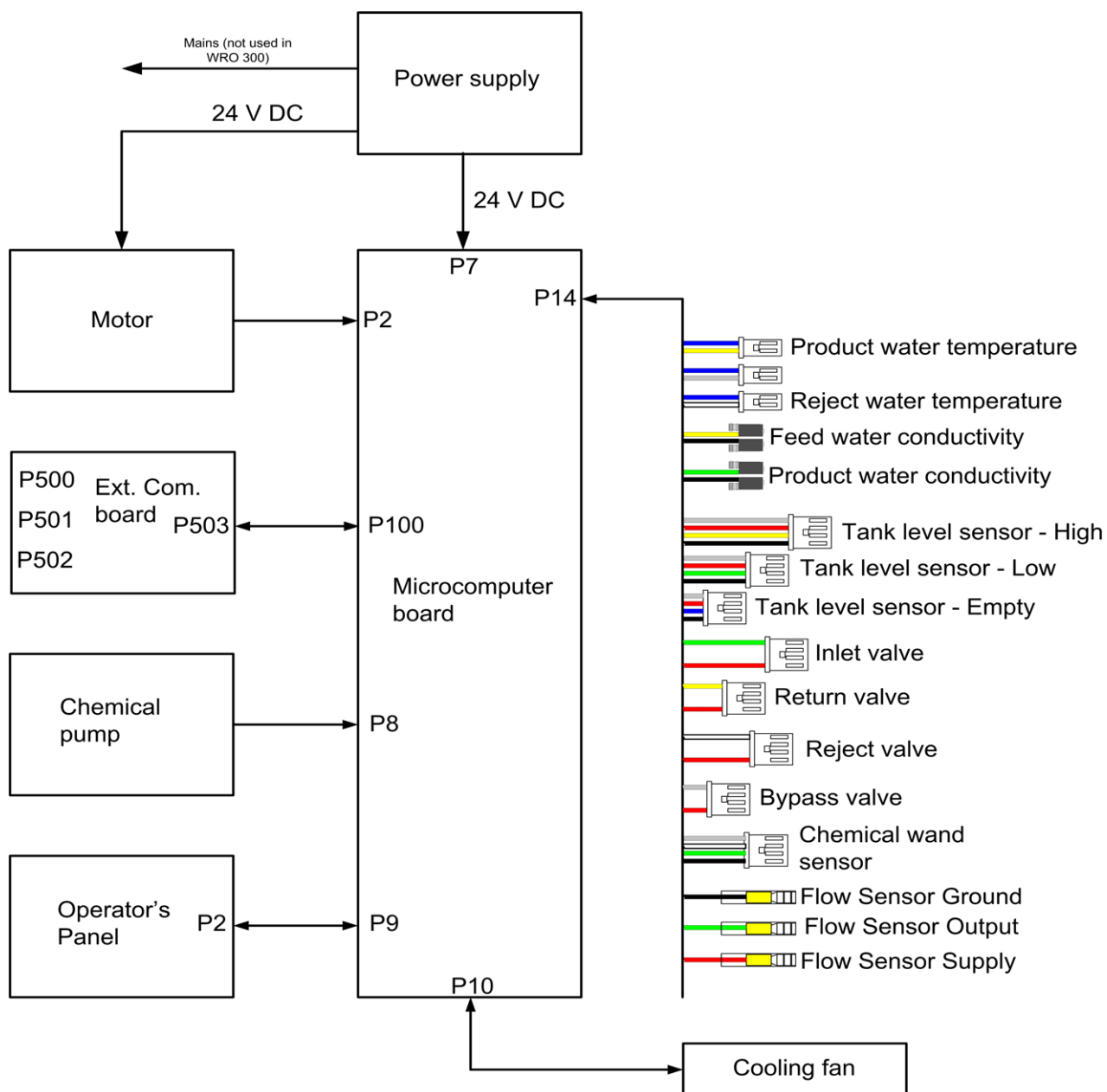


Fig.4.2 Wiring diagram for WRO 300

5 Technical data

5.1 Performance and specification

		Values	
Product water	Output flow	WRO 300	Min. 1.1 l/min at + 10 °C
		WRO 300 H	Min. 1.2 l/min at + 10 °C
	Product water loop	WRO 300	Maximum 20 meters (2 x 10m)
		WRO 300 H	Maximum 2 metres (2 x 1m)
	Product water pressure	120 to 600 kPa (17 to 87 psi) during RUN mode	
	Quality	Depends on feed water quality. If potable water is used, the following rejection rates will be obtained:	
		Total dissolved salts: > 96% (based on conductivity)	
		Bacteria and endotoxins: > 99%	
Water supply	Input	Min. 3.0 l/min required	
	Feed Water Pressur	During operation: 100 to 300 kPa (15 to 44 psi) (When there is an inlet flow of water into to the WRO)	
		Maximum design pressure: 800 kPa (116 psi)	
		If the feed water pressure exceeds 300 kPa (44 psi), a pressure regulator, order no K23389 002, shall be installed.	
	Feed Water Temperature	+ 5 to + 30 °C	
	Feed Water Quality	Potable water should be used.	
		Particle filtration shall be used < 5 µm	
		Carbon filtration shall be used if water is chlorinated, see limit below.	
		Membrane life expectancy might be reduced if the monitor is operated outside the following limits.	
	Hardness	WRO 300	< 1° dH (20 ppm as CaCO ₃) 1 grain per gallon
		WRO 300 H	< 0,3° dH (6 ppm as CaCO ₃). 0.3 grain per gallon. A softener shall be used to obtain this limit.
	Iron	< 0.1 mg/l	
	Manganese	< 0.1 mg/l	
	Turbidity	< 1 JTU	
	Total dissolved salts (TDS)	< 1500 mg/l	
	Fouling index (silt density index)	< 5	
	Chlorine (total)	< 0.1 mg/l	
	pH	2 - 11	

		Values	
Drain	Drain output flow during RUN mode	1.2 ± 0.1 l/min	
	Height difference between the drain hose outlet and the WRO drain outlet.	Max. 1.0 m	
	Drain (length of hose)	Max. 5 m	
	Drain flow capacity	Min. 3.0 l/min required	
	Max temperature WRO 300 H	85 °C	
Tank air vent	This outlet connects the tank to the atmosphere. This outlet must be connected with an air gap. If a hose is connected to the Tank air vent the hose shall have a constant slope. Note! Air will move back and forth in this line because of the variation of the level in the break tank.		
Fluid Connections	Designed for flexible, reinforced tubing 8 mm x 2.5 mm (5/16" I.D.) WRO 300 H product water loop is designed for flexible, reinforced tubing 5 mm x 3 mm		
Conductivity measurement (Temperature compensated)	Product water conductivity	Operating range	1 - 500 µS/cm
		Accuracy	± 10 % or ± 10 µS/cm
	Feed water conductivity	Operating range	10 - 2000 µS/cm
		Accuracy	± 10 % or ± 10 µS/cm
Temperature measurement	Operating range	1 - 100 °C	
	Accuracy	± 2,5 °C	
Reverse osmosis module	Membrane material	Polyamide, thin film composite	
	Membrane configuration	Spiral wound	
Power supply	Mains Voltage	100 to 240 V AC (line voltage range 85 to 264 V AC)	
	Frequency	50 or 60 Hz (line frequency range 45 to 65 Hz)	
		NOTE The user must verify the quality of the protective earth in the installation.	
	Power Consumption Max	WRO 300	570 W
		WRO 300 H	1500 W (100 V AC) 1380 W (115 V AC) 1850 W (230 V AC)
	Appliance inlet	According to IEC 60320; C14	
	Cable	Cable with protective earth Conductor length max. 3,5 m minimum 1 mm² (220-240 V AC) AWG 16 at 100 and 115 V AC	
	Mains plug	Continental Europe - 230 V AC, mains plug CEE-7 / VII United Kingdom - 230 V AC, mains plug BS 1363/A Australia - 240 V AC, mains plug AS/NZS 3112 North America type - 100 and 115 V AC, Hospital grade, earthed plug, type IEC 60 083; A5-15	
	Fuses	WRO 300	Not used
		WRO 300 H	10 A slow-blow (220-240 V AC)
			15 A slow-blow (100 and 115 V AC)
	Earth Leakage Current	Max. 250 µA (230 V AC) Max. 145 µA (115 V AC) Max. 140 µA (100 V AC)	

		Values	
Connection of external equipment	External connector 8 pins REDEL, yellow.		
	Remote start (Not available in the USA)	Max voltage	5 V DC
		Max current	100 mA DC
	External alarm	Max voltage	24 V AC or DC
		Max current	100 mA AC or DC
	External connector 8 pins REDEL, black. RS-232		
	Logging interface	Max input voltage	±15 V DC
		High level min output voltage	5.0 V DC
		Low level max output voltage	5.0 V DC
		Max Current	±5 mA DC
	External connector 8 pins REDEL, green.		
	External equipment	Accessory equipment connected to the WRO should comply with relevant standards (e.g. IEC 60950 for data processing equipment). Furthermore, medical systems should comply with IEC 60601-1-1. One important issue to consider, when configuring a system, is that the resulting leakage current for the system does not exceed acceptable limits.	

5.2 Disinfection and cleaning

Approved disinfectants	The following disinfectants may be administered via the chemical intake.	
	- Dialox, undiluted (Not available for sale in the USA) - Minncare® Cold Sterilant or Renalin 100 - Other peracetic acid disinfectants, provided that they are approved by the manufacturer for disinfection of thin film composite reverse osmosis membranes made of modified polyamide. Follow the manufacturer's instructions for Use for the specific disinfectant.	
Cleaning agents	Citric acid Other acid membrane cleaner used according to manufacturer's instructions	
Exterior cleaning	Ethanol (70%) or isopropanol (60%).	

5.3 Physical data

Measurements	Values
Weight	29 kg (WRO 300) 33 kg (WRO 300 H)
Height	563 mm
Depth	Max 520 mm
	Footprint 380 mm
Width	Max 205 mm
	Footprint 185 mm
Internal fluid volume	Approximately 4,5 litres

5.3.1 Materials in contact with product water

Type	Material
Polymers	PA (Polyamide) PEEK (Polyetheretherketone) PEI (Polyetherimide) PE (Polyethylene) POM (Polyoxymethylene) PP (Polypropylene) PPS (Polyphenylsulphite) PVC (Polyvinylchloride) PVDF (Polyvinylidenfluorid)
Rubber	EPDM (Ethylene-Propylene) NBR (Nitrile) Q (Silicone)
Metals	Titanium Stainless steel
Others	Ceramic

5.4 Environmental data

		Value
Operation	Ambient Temperature range	+ 10 to + 40 °C
	Relative Humidity range	30 to 85 % RH
	Air Pressure range	700 to 1060 hPa
Transportation and storage	Ambient Temperature range	- 10 to + 40 °C
	Relative Humidity range	10 to 95% RH
	Air Pressure range	500 to 1060 hPa
	For transportation and storage below the freezing point, the WRO must be filled with a preservation solution of 1 % formalin at least 9 % glycerol and the remaining parts RO-water.	
	During transportation and storage the equipment must be kept in its original packing. If transportation or storage time is more than 15 weeks, the environmental data relating to the operation must be followed.	
	If condensation occurs when moving the equipment between locations with different temperatures and high relative humidity (e.g. outdoor and indoor locations), the inside of the equipment must be allowed to dry before switching on the equipment.	

5.4.1 Electromagnetic environment

		Value	
The WRO is intended for use in the electromagnetic environment specified below. The customer or the user of the WRO should assure that it is used in such an environment.			
Emissions test	Compliance	Electromagnetic environment - guidance	
RF emissions CISPR 11	Group1	The WRO uses RF energy only for its internal function. Therefore its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment	
RF emissions CISPR 11	Class B	The WRO is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes	
Harmonic emissions IEC 61000-3-2	Class A (Not applicable for 115V version)		
Voltage fluctuations/flicker emissions IEC 61000-3-3	Complies (Not applicable for 115 V version)		
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance
Electrostatic discharge (ESD)IEC 61000-4-2	±6 kV contact ±8 kV Air	±6 kV contact ±8 kV Air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Surge IEC 61000-4-5	±1kV for differential mode ±2kV for common mode	1kV for differential mode ±2kV for common mode	Mains power quality should be that of a typical commercial or hospital environment
Voltage dips, short interruptions and voltage variations on power supply input lines. IEC 61000-4-11	<5% U_T ^a (>95% dip in U_T) for 0,5 cycle	<5% U_T (>95% dip in U_T) for 0,5 cycle	Mains power quality should be that of a typical commercial or hospital environment.
	40% U_T (60% dip in U_T) for 5 cycles	40% U_T (60% dip in U_T) for 5 cycles	
	70% U_T (30% dip in U_T) for 25 cycles	70% U_T (30% dip in U_T) for 25 cycles	
	<5% U_T (>95% dip in U_T) for 5 sec	<5% U_T (>95% dip in U_T) for 5 sec	

		Value	
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment-guidance^b
Conducted RFIEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 V	Portable and mobile RF communications equipment should be used no closer to any part of the WRO, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d = 1.2 \sqrt{P}$
Radiated RFIEC 61000-4-3	3 V/m 80 MHz to 3,0 GHz	3 V/m	$d = 1.2 \sqrt{P}$ 80 to 800 MHz ^c $d = 2.3 \sqrt{P}$ 800 MHz to 2,5 GHz
Radiated RF mobile phones	-	30 V/m	where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey ^d , should be less than the compliance level in each frequency range ^e . Interference may occur in the vicinity of equipment marked with the following symbol: 

Recommended separation distances between portable and mobile RF communications equipment and the WRO

The WRO is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the WRO can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the WRO as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter (m)		
	150 kHz to 80 MHz $d = \left[\frac{3.5}{3} \right] \sqrt{P}$	80 MHz to 800 MHz ^f $d = \left[\frac{3.5}{3} \right] \sqrt{P}$	800 MHz to 3 GHz $d = \left[\frac{7}{3} \right] \sqrt{P}$
0,01	0.11	0.11	0.23
0,1	0.37	0.37	0.74
1	1.2	1.2	2.3
10	3.7	3.7	7.4
100	12	12	23
Rated maximum output power of mobile phone	-	-	$d = \left[\frac{7}{30} \right] \sqrt{P}$
2W GSM/3G	-	-	0.33

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

- a. **NOTE:** U_T is the AC mains voltage prior to application of the test level.

- b. **NOTE:** These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.
- c. **NOTE:** At 80 MHz and 800 MHz, the higher frequency range applies.
- d. Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the WRO is used exceeds the applicable RF compliance level above, the WRO 300 should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the WRO.
- e. Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.
- f. **NOTE:** At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

5.5 Safety

The machine complies with the following standards:
IEC 60601-1 General requirements for safety, Class I, type B, and IPX1.
IEC 60601-1-2 Electromagnetic compatibility.

6 Troubleshooting

6.1 Alarms and notifications

General

No.	Alarm and notifications	Definition	
101	CONDUCTIVITY SENSOR FAILURE	WRO	Buzzer sounds,  flash, alarm info text after confirmation.
		Reason	The initial conductivity system check failed. If the feed water has very low conductivity the test may fail.
		Action	Check connections to conductivity sensor Replace the CPU board Check conductivity sensor calibration.
102	HIGH LEVEL SENSOR FAILURE	WRO	WRO stops, buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Failure of High level sensor in tank.
		Action	1.Check connections to tank High level sensor 2.Replace the High level sensor 3. Replace the CPU board. ^a
103	: N INTERNAL ERROR	WRO	WRO stops, buzzer sounds,  lights up, alarm info text after confirmation.
		Reason	Control unit failure.
		Action	The number (N) shown before the alarm text identifies which specific kind of alarm that has occurred. For description refer to list below. Restart unit

- a. If the casing top is removed during service external light may disturb the function of the level sensors.

List of Internal Error

No	Reason for internal error	Action
1	The alarm is issued if the system has to re-generate a new preset storage file. All preset values will be re-initialized to default values.	Restart WRO, check that no alarm has been issued, and confirm the preset values using GXP.
2	The alarm is issued if the storage file has the wrong Checksum. The file will be re-initialized with default data and the checksum will be re-calculated.	Restart WRO, check that no alarm has been issued, and confirm the preset values using GXP.
3	Internal error in the preset manager program occurring during the analysis of the preset data file. All remaining items in the preset data file will be re-initialized with default values.	Restart WRO, check that no alarm has been issued, and confirm the preset values using GXP.
4	Each item stored in the preset data area has a predefined type. The order of items in both the preset area and the default data area must be the same. If the default area is changed either an item is removed or the type is changed will force a total default re-initialization of the preset data file.	Restart WRO, check that no alarm has been issued, and confirm the preset values using GXP.

No	Reason for internal error	Action
5	The system has identified that the default-preset structure has more items than the preset file. The new preset will be added to the file, the items will be initialized with the default value. The alarm is given as indication that the preset area is changed at the start of a new program version with new presets.	Restart WRO, check that no alarm has been issued, and confirm the preset values using GXP.
6	Heater start-up test in Heat disinfection or Low flow heat. Test for no flow failed, either Heater B (product water) or Heater A (Reject water). Test time 30 seconds.	1.Check connections to the reject flow indicator, and product flow meter. Refer to "WRO 300" on page 4:2 . 2.Replace reject flow indicator and product flow meter. 3. Replace AC relay board and CPU board.
7	Heater start-up test in Heat disinfection or Low flow heat. Heater B (Product water) and/or Heater A (Reject water) detect heat during no flow test. Five seconds test time.	1.Replace reject flow indicator and product flow meter. 2.Replace AC relay board and CPU board.
8	Heater start-up test in Heat disinfection or Low flow heat. Could not establish a stable flow in reject and product water line for 20 seconds within 5 minutes.	1.Check for air leakage in fluid path. 2.Check connection between CPU board and AC relay board. 3.Replace reject flow indicator and product flow meter. 4.Replace AC relay board and CPU board.
9	Heater start-up test in Heat disinfection or Low flow heat. Could not turn on heaters when stable flow was reached. Five seconds test time.	1.Check fuses in power supply unit 2.Replace AC relay board.
10	Heater start-up test in Heat disinfection or Low flow heat. Product temperature (T1) circuit broken; too low or too high temperature detected.	1.Check connection to product flow meter. 2.Replace CPU board.
11	Heater start-up test in Heat disinfection or Low flow heat. Reject temperature (T2) circuit broken; too low or too high temperature detected.	1.Check connection to reject flow indicator. 2.Replace AC relay board.
12	No valid safe-memory bank. New memory bank created all values are set to default.	If the alarm persists replace CPU board.
13	Error when read old preset values moving to E2-prom.	Check conductivity sensors. Run rinse if required.
14	Conductivity calibration storage error.	1.Check conductivity sensors. 2.If the alarm persists replace CPU board.
15	Heater start-up test in Heat disinfection or Low flow heat. Product temperature sensor (T1) short circuit or not connected.	1.Check connection to the temperature sensor. 2.Replace CPU board. 3.Replace the temperature sensor.
16	Heater start-up test in Heat disinfection or Low flow heat. Reject temperature sensor (T2) short circuit or not connected.	1.Check connection to the temperature sensor. 2.Replace CPU board. 3.Replace the temperature sensor.
17	Heater start-up test in Heat disinfection or Low flow heat. Return temperature sensor (T3) short circuit or not connected.	1.Check connection to the temperature sensor. 2.Replace CPU board. 3.Replace the temperature sensor.
18	Mains power is detected when the Heater rods are turned on by the flow indicator relays and turned off by the control relays.	Replace AC relay board.
19	Mains power is not detected when turned on by the flow indicator relays and control relays.	1.Check fuses in the power supply unit. 2.Replace AC relay board.
20	Mains power is detected when the flow indicator relays are disabled by the CPU.	Replace AC relay board.

No	Reason for internal error	Action
21	The alarm is issued if the Heat disinfection temperature in the Heater A (Reject water) is not reached within 60 minutes.	1.Check fuses in the power supply unit 2.Check the Heater rod connections. 3.Check the preset: CHC_SUPPLY_VOLTAGE. 4.Replace Heater A (Reject water).
22	The alarm is issued if the temperature in the product water (T1) is below the Low flow heat limit for more than 15 minutes.	1.Check fuses in the power supply unit 2.Check the preset: CHC_SUPPLY_VOLTAGE. 3.Replace the Heater B (Product water).
23	The alarm is issued if it is a new machine or when the run-time data is corrupted at program start-up. The run-time will be set to zero.	If the alarm persists replace the CPU board.

May be object for changes without notice

6.1.1 Operation

No.	Alarm and notifications	Definition	
201	CONDUCTIVITY NOTIFICATION	WRO	Entering Standby mode, buzzer sounds,  flash, info text after confirmation.
		Reason	Limit for elevated conductivity has been exceeded during operation.
		Action	1.Check the RO module. 2.Check the product conductivity sensor (C2). 3.Replace the CPU board. 4.Replace the product conductivity sensor (C2).
202	CONDUCTIVITY ALARM	WRO	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Limit for high conductivity has been exceeded.
		Action	Read conductivity values 1.Check the RO module. 2.Check the product conductivity sensor (C2). 3.Replace the CPU board. 4.Replace the product conductivity sensor (C2).
203	CONDUCTIVITY STOP	WRO	WRO 300 stops, buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Maximum safety limit for Conductivity has been exceeded.
		Action	Press  until the WRO restarts (three seconds). If problem persists discontinue dialysis. 1.Check the RO module. 2.Check the product conductivity sensor (C2). 3 Replace the CPU board. 4.Replace the product conductivity sensor (C2).
205	LOW REJECTION RATE	WRO	Entering Standby mode, buzzer sounds,  flash, info text after confirmation.
		Reason	Decreased membrane performance.
		Action	1.Check the RO module. 2.Check the product conductivity sensor (C2). 3.Check the feed conductivity sensor (C1). 4.Replace the CPU board. 5.Replace the product conductivity sensor (C2). 6.Replace the feed water conductivity sensor (C1).

6.1.2 Heat and Low flow heat (only WRO 300 H)

No.	Alarm and notifications	Definition	
301	INSUFFICIENT HEAT	WRO 300 H	WRO 300 H stops, buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Power failure
		Action	Re-start the Heat disinfection cycle.
302	INSUFFICIENT WATER SUPPLY	WRO 300 H	WRO 300 H stops, buzzer sounds,  flash, info text after confirmation.
		Reason	Low level sensor indicates low level in the tank during the filling phase.
		Action	1.Check the feed water supply. 2.Check connections and the WRO leakage. 3.Check the Low level sensor and connection. 4.Check the inlet valve.
303	OVERDUE HEAT	WRO 300 H	After the higher prioritized activity, for example RUN mode, is finished and the WRO 300 H goes to Standby the buzzer sounds,  flash, info text after confirmation.
		Reason	Heat disinfection procedure is activated automatically by time channel settings but cannot be started due to other higher prioritized activity.
		Action	Check time channel settings.
304	INCORRECT WATER LEVEL	WRO 300 H	WRO 300 H stops, buzzer sounds,  and  flash, alarm info text after confirmation.
		Reason	Low level sensor indicates low level in the tank at a stage when the level is supposed to be stable.
		Action	1.Check that the dialysis machine is disconnected 2.Check fluid connections and the WRO for leakage. 3.Check the Low level sensor and connection.

6.1.3 Chem/Rinse

No.	Alarm and notifications	Definition	
401	REMOVE WAND	WRO	Buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Wand connector not removed after completion of Chemical intake.
		Action	1.Check chemical intake sensor connection. 2.Replace chemical intake sensor. 3.Replace CPU board.
402	INSUFFICIENT WATER SUPPLY	WRO	WRO stops, buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Low level sensor indicates low level in the tank during the filling phase.
		Action	1.Check the feed water supply. 2.Check connections and the WRO for leakage. 3.Check the Low level sensor and connection. 4.Check the inlet valve.
403	HIGH CONDUCTIVITY RINSE	WRO	Buzzer sounds,  flashes, alarm info text after confirmation.
		Reason	Elevated product water conductivity during final part of rinse phase.
		Action	1.Re-calibrate the product conductivity sensor (C2). 2.Replace the CPU board. 3.Replace the product conductivity sensor (C2).
404	RINSE REQUIRED	WRO	Buzzer sounds,  and  flash, alarm info text after confirmation.
		Reason	The WRO filled with disinfectant.
		Action	If the alarm persist after a second rinse cycle replace the CPU board.
405	INCORRECT WATER LEVEL	WRO	WRO stops, buzzer sounds,  and  flash, alarm info text after confirmation.
		Reason	Low level sensor indicates low level in the tank at a stage when the level is supposed to be stable.
		Action	1.Check that the dialysis machine is disconnected 2.Check fluid connections and the WRO for leakage. 3.Check the Low level sensor and connection.

6.1.4 Preservation

No.	Alarm and notifications	Definition	
501	REMOVE WAND	WRO	Buzzer sounds,  flash, alarm info text after confirmation.
		Reason	Wand connector not removed after completion of Chemical intake.
		Action	1.Check chemical intake sensor connection. 2.Replace chemical intake sensor. 3.Replace CPU board.

6.2 Boot loader

No.	Alarm and notifications	Definition	
	LOAD APPLICATION FAILED	WRO	LOAD APPLICATION FAILED is written on the display
		Reason	No main program loaded on the WRO, or corrupted program.
		Action	1.Download new program to the WRO. 2.If the LOAD APPLICATION FAILED persists, replace the CPU board.

6.3 Testing the operator's panel

- By pressing the  button during power on the program checksum status is displayed
- By pressing the  button during power on the WRO goes into program download mode and display the internal boot loader program version. A program can be downloaded from a PC with GWD software.

Following checks of the operator's panel can be done in this mode:

Button	Check indication
Press 	Display backlight flash
Press 	Buzzer signal is activated
Press 	 indication turns on
Press 	 indication turns off
Press 	 indication turns on

To exit turn the power off and on again.

7 Symbol description

NOTE

All symbols in this list may not be represented on this product.

7.1 Symbols

Symbol	Description
	Alternating current
	Protective earth (ground)
	Warning, consult accompanying documents
	Off (power, disconnection from the mains)
	On (power, connection to the mains)
	Type B, applied part
	Handle with care
	This way up
	Keep dry
	Separate collection for electrical and electronic equipment
	Input /Output
	Year of manufacturing
	Heater power supply outlet
	Conforms to requirements in EC Council Directive 93/42/EEC concerning medical devices. 0086 identifies BSI, British Standards Institution, as Notified Body.
	CSA approved for Canada

Symbol	Description
	CSA approved for Canada and USA
	CSA-listed label manufacturer
	Reject water connection
	Feed water inlet connection
	Loop connections (Product water outlet and return)

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